Too Few to Fail? Stock Market Concentration and Systemic Vulnerability

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October 24, 2026

Abstract

Stock market concentration has surged to record levels, exceeding those of the dot-com bubble and raising growing concerns among investors and regulators. This paper provides the first cross-country evidence on the effect of stock market concentration on systemic financial crises. Using a panel of 30 major stock markets from 1998 to 2020 and systemic crisis episodes from the Global Crises dataset compiled by the Behavioral Finance and Financial Stability (BFFS) project at Harvard Business School, I estimate probit models with correlated random effects (Mundlak correction) and implement a control-function approach that accounts for the predicted component of concentration using its own dynamics and observed predetermined factors, partially addressing endogeneity concerns. The results show that higher concentration significantly increases the likelihood of systemic crises, even after controlling for established macro-financial predictors.Market capitalization and credit depth emerge as the key transmission channels, with the destabilizing effect of concentration amplifying as equity markets deepen and domestic credit expands, while the liquidity channel does not generate robust amplification effects at empirically observed levels of market activity. Robustness checks, including alternative crisis definitions, functional forms, and alternative first-stage specifications, confirm the stability of the findings. The evidence suggests that stock market concentration is not merely a byproduct of financial development, but an important feature associated with systemic fragility, with potential implications for the design of macroprudential policy.

Keywords: Stock Market Concentration; Systemic Crisis; Financial Stability; Market Structure

JEL classification: G01; G15; G18; C23.

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1 Introduction

Stock market concentration has intensified in recent years, approaching record highs. As of August 2024, the ten largest firms accounted for 34% of the S&P 500's capitalization, compared with about 27% at the peak of the dot-com bubble in 2000 (Capital Group, 2024[1]; CFA Institute, 2025[2]). Unlike the 1990s, when concentration reflected speculative valuations of unproven firms, today's dominance stems from the entrenched profitability and global reach of a small group of firms[3], amplified by the rise of passive investment flows. This shift has made aggregate market performance increasingly dependent on the fortunes of just a few firms, raising the question of whether equity market structure itself has become a source of systemic fragility [4].

In principle, equity markets promote stability by dispersing risk across firms and sectors through diversification, liquidity, and price discovery. From this perspective, the rise of dominant firms may simply reflect efficiency gains, economies of scale, technological leadership, and global reach, that strengthen aggregate performance. Analogies can be drawn to banking, where concentrated systems have sometimes been argued to enhance resilience by allowing large institutions to diversify more effectively and absorb losses (Allen and Gale, 2000; Beck et al., 2003).

Yet an opposing view emphasizes that excessive concentration can itself be a source of fragility. Recent research highlights that the surge in stock market concentration is not only a byproduct of firm fundamentals but also of structural shifts in asset management. In particular, Jiang et al. (2024), show that flows into passive funds disproportionately raise the prices of the economy's largest firms, especially those already overvalued, thereby reinforcing their market share. This

[1]Stern, E., et al. 2024. "Has Stock Market Concentration Reached a Tipping Point?", *Capital Group.*

[2]Pauley, B., Bales, K., & Schreiber, A. (2025). "Market Concentration and the Lost Decades.", *CFA Blog.*

[3]Lakos-Bujas, D. (2024). "Stock Market Concentration Is on the Rise. Will This Continue?", *J.P. Morgan Insights.*

[4]Lamont, O., et al. (2024). "Market Concentration: How Big a Worry?", *Goldman Sachs.*

structural force strengthens the case that concentration is not transitory but a persistent feature of modern equity markets, with important implications for systemic risk. When market capitalization is narrowly distributed, the system becomes more exposed to the performance of a few firms, magnifying the impact of idiosyncratic shocks. Evidence at the micro level shows how fire sales triggered by investor redemptions depress prices and transmit stress across otherwise unrelated assets (Coval and Stafford, 2007). Ownership concentration and correlated trading exacerbate fragility by amplifying volatility in downturns (Greenwood and Thesmar, 2011). Similar mechanisms operate in banking, where interconnected balance sheets and concentrated exposures make large institutions "too interconnected to fail" (Greenwood et al., 2015). In equity markets, high concentration may interact with passive flows and index rebalancing to create channels through which distress at the top propagates system-wide.

Cross-country and macro-level evidence reinforces these concerns. Stock market concentration has been linked to slower innovation and weaker dynamism (Bae et al., 2021; Neuhann and Sockin, 2024; Zhang et al., 2025), while firm-level studies show that larger, more leveraged companies suffer disproportionate losses during crashes (Wang et al., 2009). At the global level, increased financial integration has further reduced diversification benefits and accelerated contagion across markets (Lehkonen, 2015). Theoretical research also provides insight into the mechanisms through which financial systems become vulnerable to shocks. Song Shin (2005) emphasizes how liquidity constraints and balance-sheet feedbacks can generate self-reinforcing downturns, producing "twin crises" through procyclical amplification. Danielsson et al. (2013) develop the concept of endogenous risk, showing how systemic fragility can arise endogenously through leverage, fire sales, and correlated trading even in the absence of large external shocks. Although these studies do not examine stock market concentration directly, they establish that systemic fragility can emerge

endogenously from structural features of financial systems.

Taken together, these findings suggest that the relationship between concentration and stability is theoretically and empirically ambiguous. On the one hand, large and diversified firms may provide stability by absorbing shocks. On the other hand, concentration can magnify fragility by tying the system's performance to a narrow set of firms and increasing the likelihood that shocks propagate through correlated portfolios and herding dynamics. Despite growing recognition of these channels, no study has systematically examined the direct link between stock market concentration and the incidence of systemic financial crises across countries. This paper is the first to provide evidence of the effect stock market concentration on the probability of systemic crises. Using panel data from 30 stock exchanges across 30 countries between 1998 and 2020, I provide cross-country evidence that equity market concentration is strongly and robustly associated with systemic fragility, even after controlling for established macro-financial predictors. Stock market concentration is measured as the share of total market capitalization accounted for by the ten largest firms, using data from the World Federation of Exchanges. Crises are identified primarily using the Behavioral Finance and Financial Stability (BFFS) database compiled by Reinhart and Rogoff (2009), with the Laeven and Valencia (2020) dataset employed both as a complement and as an independent robustness check. The baseline specification is a panel probit model that estimates the probability of crisis as a function of stock market concentration and standard macro-financial controls. To account for unobserved heterogeneity, I employ the Mundlak correction, which allows the model to capture within-country variation while retaining the advantages of a random-effects framework (Wooldridge, 2010).

To model the dynamic structure of stock market concentration and assess the sensitivity of the results to potential endogeneity, I implement a control-function approach following the residual

inclusion framework for nonlinear models (Wooldridge, 2015). In the first stage, concentration is modeled as a function of its own lag and observed predetermined factors, capturing the persistent and structurally determined component of equity market concentration. The generalized residual from this equation, which represents the unanticipated or innovation component of concentration, is then included in the second-stage crisis probit. This procedure serves two purposes. First, it separates the predicted, slowly moving component of concentration from its transitory innovation, allowing the crisis equation to reflect the dynamic evolution of market structure. Second, it provides a Hausman-type test of endogeneity: a significant coefficient on the residual indicates that the baseline specification without it would be inconsistent. Across all specifications, the residual enters significantly, confirming the presence of endogeneity. Importantly, the coefficient on stock market concentration remains large, positive, and highly significant after the inclusion of the residual, indicating that the strong predictive relationship between concentration and systemic crises is robust to accounting for both the dynamic process governing concentration and the endogenous component captured by the first-stage predetermined factors. The channel analysis shows that the destabilizing effect of concentration strengthens as equity markets deepen, while concentration raises crisis risk across the full range of credit conditions without systematically interacting with credit depth. The evidence for a liquidity channel, emphasized in theories of fire sales and amplification such as Song Shin (2005) and Danielsson et al. (2013), is inconclusive at empirically representative levels of market activity.

To ensure robustness, I validate the baseline results using alternative functional forms: a logit model, a complementary log-log (cloglog) model appropriate for rare events, and a linear probability model (LPM) that, while less suitable econometrically, provides an intuitive benchmark. Across all models, I compute average marginal effects to translate coefficients into economically

meaningful magnitudes.

This paper makes three contributions to the financial stability literature. First, it is the first to document a systematic relationship between equity market concentration and systemic crises, complementing existing work on credit booms, bubbles, and banking fragility. Second, it connects insights from theories of liquidity spirals, endogenous risk, and fire sales with cross-country evidence on systemic crises. Third, it underscores the importance of monitoring equity market structure in macroprudential surveillance, particularly in light of the unprecedented levels of concentration observed in today's global markets.

The remainder of the paper is structured as follows. Section 2 positions the paper within the related literature on financial crises and market concentration. Section 3 describes the data and outlines the construction of the stock market concentration measure and systemic crisis indicators. Section 4 details the empirical framework, including the panel probit specification, the Mundlak correction, and the control-function approach. Section 5 presents the main results and the channel analysis, which shows that the effect of concentration strengthens with market depth, while also examining alternative specifications of the baseline model. Section 6 reports a series of robustness tests, including alternative functional forms and crisis definitions. Section 7 concludes by discussing the broader implications for systemic risk monitoring and macroprudential policy.

2 Literature Review

The early literature on financial crises emphasizes macroeconomic imbalances, credit booms, current account deficits, asset bubbles, and fiscal excesses. Reinhart and Rogoff (2009) document that crises typically involve deep and prolonged contractions in output, asset prices, and employment,

often requiring major policy interventions. Laeven and Valencia (2020) formalize this understanding with a comprehensive cross-country database of systemic banking crises, which constitutes a central resource for empirical research. Empirical studies consistently highlight macro-financial predictors of crises. Davis et al. (2016) use panel logit and linear probability models for 35 countries (1970–2010) and show that rapid private credit growth combined with deteriorating current accounts strongly raises crisis risk. Roy (2022) emphasizes housing bubbles and external deficits in advanced economies as dominant precursors of instability. Shen and Hsu (2022) introduce a panel threshold logit model, showing that crisis probabilities increase sharply once credit-to-GDP or interest rates cross critical thresholds. This evidence underscores that systemic crises are often preceded by sustained macroeconomic imbalances.

Building on this macro perspective, a second strand of work explores how financial market structure contributes to systemic fragility. Banking concentration has received particular attention. Beck et al. (2006) and Bretschger et al. (2012) provide cross-country evidence that higher banking concentration can destabilize financial systems, especially in low-income economies with weaker institutions. Greenwood et al. (2015) highlight how concentration and interconnectedness render banks vulnerable, with small shocks amplified through balance-sheet contagion. These debates reflect two competing views: the "concentration–stability" hypothesis, which emphasizes diversification, and the "concentration–fragility" hypothesis, which stresses systemic exposure.

Analogous concerns have been raised in equity markets. A key strand of literature highlights amplification mechanisms under stress. Coval and Stafford (2007) show that fire sales caused by investor redemptions can depress asset prices and spill over to unrelated securities. Greenwood and Thesmar (2011) develop the concept of "fragility" in concentrated ownership structures, demonstrating that correlated trading amplifies volatility during downturns. Patro et al. (2013) show

that idiosyncratic return correlations spike during crises, reflecting systemic herding and a breakdown of diversification. At the firm level, Wang et al. (2009) and Andreou et al. (2021) find that smaller, more leveraged, and opaque firms experience disproportionately severe losses in crises. In highly concentrated markets, their limited weight reduces the scope for systemic diversification, leaving aggregate outcomes more dependent on a small number of large firms. As a result, concentration and correlated exposures amplify fragility in periods of stress.

The literature has also linked stock market concentration to innovation and capital allocation. Bae et al. (2021) document that when the top five firms account for a larger share of market capitalization, IPO activity, firm entry, R&D, and productivity growth all decline. Neuhann and Sockin (2024) show theoretically and empirically that concentration misallocates capital: dominant firms attract flows even when less productive, crowding out more efficient competitors and increasing systemic vulnerability. A complementary strand of research focuses on the role of passive investment flows (Jiang et al., 2024) demonstrate that passive investing systematically channels disproportionate demand into the largest firms, raising their valuations and increasing their idiosyncratic volatility. This mechanism provides a structural explanation for the sustained rise in concentration, linking asset management trends to the fragility channels emphasized in this paper.

Dynamic studies emphasize that crises themselves can reinforce concentration. Zhang et al. (2023) find that post-crisis recoveries are slower for small- and mid-cap firms, which deepens concentration after shocks. Lehkonen (2015) shows that equity market co-movement increased sharply during the 2008 global financial crisis, especially in developed markets, reducing international diversification benefits precisely when they were needed most. This suggests that crises not only emerge from concentrated structures but also reshape market concentration in their aftermath.

Theoretical contributions also provide insight into how concentration can amplify systemic risk.

Song Shin (2005) emphasizes liquidity spirals and balance-sheet effects, showing how procyclical feedbacks can trigger twin crises in banking and currency markets. Danielsson et al. (2013) introduce the concept of endogenous risk, where systemic instability emerges from within the financial system through leverage, fire sales, and correlated behavior. These frameworks imply that equity market concentration can serve as a structural amplifier of endogenous risk, magnifying systemic fragility when shocks affect dominant firms.

In summary, the literature has advanced in two broad directions: one focusing on macro-financial imbalances (credit booms, bubbles, and external balances) as crisis predictors, and another on market structure and concentration as amplifiers of fragility. While existing evidence shows that concentration impairs innovation, distorts capital allocation, and magnifies volatility through fire sales and correlated trading, its relationship with systemic financial crises remains relatively underexplored. This paper is the first to bridge these strands by providing cross-country empirical evidence that stock market concentration significantly increases the probability of systemic crises, thereby extending the financial stability literature to a novel and previously overlooked dimension of market structure.

3 Methodology

3.1 Data

This study investigates whether stock market concentration influences the probability of systemic financial crises using an unbalanced panel of 30 countries from 1998 to 2020.[5] Systemic crisis

[5]The country sample for the regression analysis using the BFFS definition of systemic crises includes both advanced and emerging economies: Australia, Austria, Brazil, Canada, Chile, Colombia, Germany, Switzerland, Spain, the United Kingdom, Greece, Ireland, Italy, Japan, South Korea, Morocco, Mexico, Malaysia, Norway, New

episodes are identified using two complementary sources. The baseline crisis variable follows Reinhart and Rogoff (2009), as compiled in the Harvard BFFS dataset. In their framework, a systemic crisis is defined as a period of widespread banking distress, characterized by bank runs, failures, or major liquidations, typically requiring extensive government intervention such as recapitalizations, deposit freezes, or nationalizations of large institutions. This broad definition emphasizes the macroeconomic impact and prolonged aftermath of financial distress.

To test robustness, I construct an alternative crisis variable based on Laeven and Valencia (2020). Their definition is narrower and more operational: a systemic crisis is recorded only when two conditions are jointly satisfied—(i) the banking system exhibits significant distress (bank runs, large losses, or widespread liquidations), and (ii) the authorities undertake major policy interventions, such as nationalizations, liquidity support exceeding 5% of deposits, guarantees, asset purchases, or fiscal restructuring costs above 3% of GDP. The onset year is coded as the first year in which both conditions are met.

The central explanatory variable is stock market concentration (*Stock Market Concentration*), measured as the market share of the top ten firms' capitalization relative to the total domestic market capitalization. This measure captures the degree of dominance by a narrow group of large firms and provides a proxy for the vulnerability of equity markets to firm-specific shocks.

Table 2 reports the descriptive statistics for the main variables. Crises are relatively rare, with systemic events occurring in 10.5% of the sample years and alternative crisis measures in 8.1%. This frequency is consistent with the view of crises as infrequent but high-impact events. Stock market concentration averages 53.1 with a standard deviation of 18.9, ranging from below 10 to

Zealand, the Philippines, Poland, Russia, Singapore, the United States, and South Africa.

Using the alternative definition of Laeven and Valencia (2020), the sample is extended to include Hong Kong, Hungary, and Iran, which are classified as experiencing systemic crises in their dataset.

nearly 100. The wide dispersion indicates that while some countries have relatively broad-based equity markets, others are almost entirely dominated by a few firms.

Domestic credit averages 88% of GDP but varies substantially, with some emerging economies under 10% and advanced financial systems above 200%. The current account averages close to balance but spans persistent deficits of 16.8% of GDP to surpluses above 27%. The financial account shows similar dispersion, ranging from outflows of 17% to inflows of 26% of GDP, consistent with the volatility of capital flows. Inflation averages 4.9%, though the distribution is highly skewed, with some crises associated with extreme spikes exceeding 100%. GDP per capita averages about $23,000, but the minimum and maximum span three orders of magnitude—from below $1,000 in low-income economies to above $110,000 in wealthy financial centers.

Market depth also varies widely: market capitalization to GDP averages 93% but ranges from under 1% to nearly 1,800%, while turnover averages 257% with some exchanges recording extraordinarily high trading intensity. Institutional variables show strong heterogeneity: creditor rights range from 0 to 4, religious fractionalization from nearly homogeneous societies to highly diverse ones, and about 36% of the sample adopts UK legal origin.

Tables 3 and 4 present country averages. Systemic crises remain rare in most economies, but Hungary, Greece and Ireland record non-trivial incidence, reflecting episodes of banking instability and sovereign debt distress. Stock market concentration displays striking cross-country patterns: in advanced, diversified markets such as the United States, Canada, and the United Kingdom, concentration averages below 30; in contrast, small or emerging economies such as Hungary[6] ,Morocco, and Colombia record concentration above 70, in some cases exceeding 90. These differences

[6]Hungary exhibits unusually high levels of stock market concentration, making it an outlier in the sample. To ensure robustness, the main models are re-estimated excluding Hungary; the results remain qualitatively unchanged.

underscore the structural contrasts between mature, liquid markets and narrow, firm-dominated ones.

GDP per capita highlights the economic heterogeneity of the sample, with high-income economies such as Switzerland, and the United States averaging above $50,000, while countries such as India, Indonesia, and the Philippines average below $5,000. Inflation follows a similar split: advanced economies like Japan and Switzerland show persistent low inflation or mild deflation, whereas emerging markets such as Brazil record sustained double-digit price growth. Current account balances also diverge structurally, with consistent surpluses in export-driven economies like Singapore and Switzerland, versus persistent deficits in the U.K., U.S., and South Africa. The financial account shows extreme movements in Hungary and other transition economies, likely reflecting privatization inflows and volatile foreign investment.

Market development indicators also reveal wide heterogeneity. The U.S., Switzerland, and Singapore display very high market capitalization relative to GDP (often above 120%), whereas many emerging economies (e.g., Morocco, Colombia, the Philippines) average below 50%. Turnover ratios likewise diverge, with highly liquid exchanges (U.S., Hong Kong, South Korea) exceeding 150%, while others (New Zealand, Morocco) remain below 20%. These country-level differences highlight the diversity of equity market structures, which is central to the empirical strategy of testing how concentration conditions systemic risk.

Table 5 reports Pearson correlations among the main variables. Several important patterns emerge. First, stock market concentration is positively correlated with systemic crises ($\rho = 0.17$), consistent with the hypothesis that more concentrated equity markets are more fragile. The correlation is moderate in size and statistically significant. Second, SMC is negatively correlated with market capitalization ($\rho = -0.19$) and creditor rights ($\rho = -0.39$). These bivariate associations

suggest that, on average, larger markets and stronger legal environments are associated with lower measured concentration. However, these simple correlations should be interpreted cautiously, as they do not account for other factors that may simultaneously influence market structure and financial stability. In the multivariate analysis presented later in the paper, the relationship between creditor rights and stock market concentration differs from the simple correlation shown here, suggesting that institutional factors may shape market structure in more complex ways. Finally, SMC is positively correlated with religious fractionalization ($\rho = 0.47$), which may partly reflect the prevalence of more concentrated equity markets in economies with greater institutional and societal heterogeneity. Third, systemic crises are negatively correlated with GDP per capita ($\rho = -0.19$) and positively with inflation ($\rho = 0.11$), consistent with the macro-financial literature linking crises to weak fundamentals and price instability. Fourth, the current and financial accounts are highly correlated ($\rho = 0.95$), which is expected given their accounting identity. Both serve as measures of external imbalances, but they capture different aspects: the current account reflects persistent trade and income flows, while the financial account captures the volatility of capital movements. In the regressions, the financial account often shows stronger explanatory power, suggesting that abrupt capital flow reversals may be more closely linked to systemic crises than gradual trade imbalances. Finally, institutional variables are mutually correlated. Legal origin is positively associated with creditor rights and negatively with religious fractionalization, capturing how common-law systems tend to coincide with stronger investor protections and lower societal fragmentation. While these institutional correlations are high, robustness checks show that collinearity does not drive the results.

In sum, the descriptive statistics underscore three points: (i) crises are rare but economically

significant events; (ii) stock market concentration varies widely across countries and is systematically higher in smaller, less developed markets; and (iii) concentration is empirically linked to both systemic crises and institutional quality. This motivates the central question of the paper: whether stock market concentration is systematically associated with the probability of systemic financial crises, even after controlling for macro-financial and institutional determinants.

3.2 Econometric Methodology

Given the binary nature of the dependent variable and the rarity of systemic crises, the baseline specification is a random effects probit model. This choice reflects the structure of the dataset. Fixed effects probit estimators, such as Chamberlain's conditional likelihood approach (Chamberlain, 1980), are subject to the incidental parameters problem and become unreliable in short panels where the dependent variable exhibits little within-country variation. In this sample, many countries experience no systemic crisis events during the observation window. Applying fixed effects would therefore eliminate entire units and discard cross-sectional variation that is central to the analysis. By contrast, the random effects framework retains all available information and permits the inclusion of time-invariant or slowly evolving regressors.

To relax the strict exogeneity assumption underlying random effects, I complement the baseline specification with a correlated random effects (CRE) model using the Mundlak (1978) correction. This approach augments the probit regression with the country-specific means of the time-varying covariates, thereby accounting for potential correlation between unobserved heterogeneity and explanatory variables. As Wooldridge (2010) notes, this strategy maintains consistency while avoiding the efficiency losses associated with fixed effects, making it particularly well-suited for

short and unbalanced macro-financial panels. Formally, I test the validity of the CRE specification using a joint Wald test of the Mundlak means. Rejection of the null confirms that unobserved heterogeneity is correlated with the regressors, implying that standard random effects would be inconsistent.The Wald statistic is large ($\chi^2 = 51.95$, $p < 0.001$; Table 17), providing strong support for the CRE specification as the baseline.

Beyond the probit specification, I test robustness across alternative binary response models. First, a logit model is estimated to verify that the results are not specific to the normal distributional assumption of the probit. Second, a complementary log-log (cloglog) model is implemented, which is particularly appropriate for rare events given its asymmetric link function that captures nonlinear hazard dynamics. Finally, a linear probability model (LPM) is estimated as a transparent benchmark. While the LPM has well-known drawbacks, including heteroskedastic errors and non-bounded predicted probabilities, it provides coefficients that are easily interpretable as marginal effects. Together, these alternative link functions serve as robustness checks against functional form misspecification, following best practice in the early warning systems literature (e.g., Jemović and Marinković, 2021).

A further concern is the potential endogeneity of stock market concentration. Concentration may itself be influenced by crisis risk, for example through reverse causality when financial distress reshapes market structure. To model the dynamic structure of stock market concentration and assess the sensitivity of the results to potential endogeneity, I implement a control-function approach. In the first stage, stock market concentration is modeled as a function of its own lag and observed predetermined factors, including creditor rights, religious fractionalization, and legal origin, capturing the persistent and structurally determined component of equity market concentration. The generalized residual from this equation is then included in the second-stage crisis probit. This

procedure provides a Hausman-type test of endogeneity: a significant coefficient on the residual indicates that the baseline specification without it would be inconsistent, while insignificance supports the baseline model. Importantly, the control-function framework accommodates nonlinear probability models and maintains interpretability in terms of average marginal effects.

Explanatory variables are lagged by one period throughout the analysis. This timing convention is standard in early warning system frameworks, reflecting the economic structure in which current macro-financial conditions affect future crisis risk (Bretschger et al., 2012). Lagging also mitigates concerns of simultaneity and contemporaneous feedback effects between crises and financial variables (Bae et al., 2021).

Finally, the analysis incorporates several dimensions of capital market structure as robustness checks. Stock market capitalization (as a share of GDP) is included to capture market depth, while stock market turnover serves as a measure of liquidity. These variables allow me to examine whether the observed relationship between concentration and systemic crises persists after accounting for broader features of market size and trading intensity (Lehkonen, 2015; Andreou et al., 2021).

Overall, this modelling strategy combines a theoretically motivated baseline with multiple diagnostic tests. The random effects probit with Mundlak correction provides a consistent framework suited to the panel structure; a Wald test of the Mundlak means formally evaluates the CRE specification; alternative link functions confirm robustness to distributional assumptions; the control-function approach both corrects for and tests endogeneity; and the inclusion of controls disentangles concentration effects from broader features of financial development. This design helps reduce concerns that the findings are driven by functional form assumptions or unobserved heterogeneity, and instead supports the interpretation that stock market concentration is systematically associated with systemic financial crises.

3.2.1 Econometric Specification

Formally, let $Y_{it} \in \{0, 1\}$ indicate the incidence of a systemic crisis in country i and year t. The latent index representation is

$$Y_{it}^{*} = \beta\, X_{it} + \boldsymbol{\gamma}'\mathbf{C}_{it-1} + u_i + \varepsilon_{it}, \tag{1}$$

$$Y_{it} = \mathbf{1}\{Y_{it}^{*} > 0\}, \tag{2}$$

where X_{it} denotes stock market concentration, $\mathbf{C}_{it}$ is a vector of controls, $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is a country-specific random effect, and $\varepsilon_{it} \sim \mathcal{N}(0, 1)$ is the idiosyncratic error. Standard errors are clustered at the country level to allow for arbitrary within-country correlation.Year fixed effects are not included in the baseline specification due to the limited sample size and the large number of parameters already estimated, including the macro-financial controls and the Mundlak correction terms, which substantially reduce the available degrees of freedom. Moreover, the macro-financial covariates included in the model, such as GDP growth, inflation, exchange rate changes, and the current and financial accounts, already capture much of the common time variation driven by global economic conditions, reducing the need for separate year effects.

Stock market concentration is potentially endogenous due to reverse causality (e.g., crises altering market structure) and omitted drivers of both concentration and systemic risk. To model the dynamic structure of concentration and assess the sensitivity of the results to potential endogeneity, I implement a control-function strategy. The first-stage equation for concentration is

$$X_{it} = \rho\, X_{i,t-1} + \boldsymbol{\pi}'\mathbf{C}_{it-1} + \boldsymbol{\delta}'\mathbf{W}_{it-1} + \eta_i + v_{it}, \tag{3}$$

where $\mathbf{W}_{it-1}$ is the full set of observed structural factors such as creditor rights, religious fractionalization, and legal origin, $\mathbf{C}_{it-1}$ all the control variables, η_i are country effects, and v_{it} is the residual innovation. Predicted residuals $\hat{v}_{it}$ are then added to the crisis equation. The likelihood contribution for individual i under the random-effects probit is

$$\mathcal{L}_i = \int \left[\prod_{t=1}^{T_i} \Phi\big(\beta X_{it} + \boldsymbol{\gamma}'\mathbf{C}_{it-1} + \lambda \hat{v}_{it} + u\big)^{Y_{it}} \Big(1 - \Phi\big(\beta X_{it} + \boldsymbol{\gamma}'\mathbf{C}_{it-1} + \lambda \hat{v}_{it} + u\big)\Big)^{1-Y_{it}} \right] \phi(u; 0, \sigma_u^2)\, du, \tag{4}$$

where $\Phi(\cdot)$ is the standard normal cdf and $\phi(u; 0, \sigma_u^2)$ is the normal density of the random effect. The unconditional probability of a crisis for observation (i, t), obtained by integrating over u_i, is

$$P(Y_{it} = 1 \mid X_{it}, \mathbf{C}_{it-1}) = \int \Phi\big(\beta X_{it} + \boldsymbol{\gamma}'\mathbf{C}_{it-1} + \lambda \hat{v}_{it} + u\big)\, \phi(u; 0, \sigma_u^2)\, du, \tag{5}$$

which serves as the basis for computing average marginal effects below. A Wald test of $H_0 : \lambda = 0$ provides a direct test of exogeneity of concentration. Significance of λ would indicate that endogeneity is present, confirming the relevance of including the control-function residual. To confirm robustness across different link functions, I estimate the model using three alternative binary response models: logit, complementary log-log (cloglog), and linear probability (LPM). The logit provides an alternative symmetric specification; the cloglog is particularly appropriate for rare events such as systemic crises; and the LPM, while less theoretically grounded, offers a transparent linear benchmark. All models report both coefficient estimates and average marginal effects, ensuring comparability of the economic magnitudes across functional forms. The crisis

equation, including all covariates used in the empirical analysis, is specified as

$$\begin{aligned}
\text{SystemicCrisis}^{*}_{i,t} = {} & \beta_0 + \beta_1 \,\text{StockMktConc}_{i,t} \\
& + \beta_2 \,\text{DomesticCredit}_{i,t-1} + \beta_3 \,\text{GDPpc}_{i,t-1} + \beta_4 \,\text{Inflation}_{i,t-1} \\
& + \beta_5 \,\text{CurrentAccount}_{i,t-1} + \beta_6 \,\text{FinAccount}_{i,t-1} \\
& + \beta_7 \,\Delta\text{Trade}_{i,t-1} + \beta_8 \,\Delta\text{Rate}_{i,t-1} \\
& + \beta_9 \,\text{StockMktCap}_{i,t-1} + \beta_{10} \,\text{StockMktTurnover}_{i,t-1} \\
& + u_i + \varepsilon_{it}, \qquad (6)
\end{aligned}$$

with $Y_{it} = \mathbf{1}\{\text{SystemicCrisis}^{*}_{i,t} > 0\}$. Here, stock market concentration enters contemporaneously as the variable of interest. Conceptually, this reflects the view that it is the prevailing structure of the equity market at the time of a crisis that conditions systemic vulnerability, as the amplification mechanisms emphasized in the literature, such as fire sales (Coval and Stafford, 2007), correlated trading and ownership fragility (Greenwood and Thesmar, 2011), and endogenous risk through leverage and balance-sheet feedbacks (Danielsson et al., 2013; Song Shin, 2005), operate through the contemporaneous distribution of market capitalization across firms. Econometrically, this timing is supported by the structure of the first-stage equation, where concentration at time t is modeled as a function of its own lag and predetermined covariates at $t-1$. The predicted value from this equation therefore represents the expected level of concentration at t given information available at $t-1$, ensuring that the contemporaneous entry of concentration in the crisis equation is based on its predetermined dynamic component rather than on simultaneous realizations. All other controls are lagged one period to ensure they are predetermined relative to the crisis outcome,

and u_i captures country-specific heterogeneity. I report *average marginal effects* (AMEs), which convert probit coefficients into changes in crisis probability. For a continuous regressor x_{kit}, the marginal effect at observation (i, t) is

$$\mathrm{ME}_{it}(x_k) = \int \phi\big(\beta X_{it} + \boldsymbol{\gamma}'\mathbf{C}_{it-1} + \lambda \hat{v}_{it} + u\big)\, \beta_k\, \phi(u; 0, \sigma_u^2)\, du,$$

where $\phi(\cdot)$ denotes the standard normal density. The AME is the sample average,

$$\overline{\mathrm{ME}}(x_k) = \frac{1}{N} \sum_{i,t} \mathrm{ME}_{it}(x_k),$$

which measures the expected change in the probability of a systemic crisis from a one–unit increase in x_k. Reporting AMEs facilitates comparability across models and makes the economic magnitude of coefficients directly interpretable.

3.2.2 Predetermined Factors in the First Stage

The first-stage equation models stock market concentration as a function of its own lag and a set of observed predetermined factors that capture the structural and institutional determinants of equity market composition. These factors include the *Creditor Rights Index* of Djankov et al. (2007), measured in 2002, religious fractionalization (Alesina et al., 2003), and legal origin (La Porta et al., 1999).

The inclusion of these variables is motivated by the law-and-finance literature, which documents that institutional characteristics shape the structure of financial markets across countries. Creditor protection, for instance, affects firms' financing choices and the breadth of public equity

markets: stronger enforcement expands and cheapens debt finance, shifting the composition of firms that access public equity. Mid-sized firms substitute toward bank finance or remain private, while large, reputable firms continue to issue in public markets. The net effect is a skewed equity market composition in which large incumbents dominate, raising measured concentration. This channel is well documented in the cross-country law-and-finance literature (Cho et al., 2014; Brockman and Unlu, 2009). Similarly, religious fractionalization and legal origin capture deeper institutional variation in ownership structures, investor protection, and the development of public equity markets (La Porta et al., 1999; McCaig and Stengos, 2005).

These variables are effectively time-invariant during the sample period, providing stable cross-country variation without introducing spurious dynamics. Across all specifications, the predetermined factors enter with statistically significant coefficients, and the corresponding regression explains a substantial share of the variation in stock market concentration. This indicates that the predicted and residual components of concentration are clearly distinguished within the empirical framework.

These predetermined factors enter the first-stage concentration equation but are excluded from the crisis probit. While the law-and-finance literature has documented associations between institutional characteristics and financial fragility (Houston et al., 2010; Demirgüç-Kunt and Detragiache, 1998), these findings are based on cross-country comparisons and do not establish a direct, time-varying effect on crisis incidence. In the present setting, creditor rights, religious fractionalization, and legal origin are effectively constant throughout the sample period. They cannot, by construction, explain within-country variation in the timing of systemic crises. Any cross-country differences in crisis propensity that these institutional features might generate are absorbed by the country-specific random effects and the Mundlak correction terms, which together

capture persistent between-country heterogeneity. The remaining channels through which institutional characteristics could affect crisis risk, credit expansion, financial market depth, and bank risk-taking, are explicitly controlled through the time-varying macro-financial covariates.

4 Results

We begin our empirical investigation by exploring how stock market concentration differs between crisis and non-crisis periods. Figure 1 presents a boxplot comparing the distribution of stock market concentration (measured as the share of market capitalization held by the top 10 firms) across systemic crisis states. The median concentration is visibly higher during crisis episodes, with the interquartile range also shifting upward. The presence of long upper tails in both groups suggests heterogeneity across countries, but the bulk of observations during crisis years are clustered at substantially higher concentration levels. This pattern provides preliminary visual evidence that systemic crises tend to occur in environments where a small number of firms dominate equity markets.

[INSERT Figure 1]

This insight is reinforced by the kernel density estimates in Figure 2, which plot the smoothed distribution of stock market concentration separately for crisis and non-crisis observations. The crisis density curve (dashed orange) is noticeably right-shifted, peaking at higher concentration levels and maintaining a thicker upper tail relative to the non-crisis distribution. While both densities are unimodal, the difference in location and shape confirms that the likelihood of a crisis increases in more concentrated equity markets, even before controlling for other variables.

[INSERT Figure 2]

4.1 Baseline Results

Table 6 reports the structural estimates of the relationship between stock market concentration (SMC) and the probability of systemic crisis. Across all four specifications, probit, logit, cloglog, and linear probability model, the coefficient on SMC is positive and statistically significant at the 1% level. This finding provides strong evidence that equity markets dominated by a small set of firms are more vulnerable to systemic financial distress. The consistency of the estimates across different link functions confirms that the result does not depend on distributional assumptions.

Control variables behave in line with theoretical priors. Domestic credit is positive and significant in all specifications, underscoring the role of credit booms in amplifying systemic risk[7]. GDP per capita enters with a positive and significant coefficient, while its Mundlak mean enters negatively. These two terms capture distinct sources of variation: the positive coefficient on the time-varying GDP per capita reflects short-run deviations from a country's long-run average, suggesting that periods of above-trend income growth are associated with elevated crisis risk, consistent with overheating dynamics and procyclical financial vulnerabilities. The negative Mundlak mean, by contrast, captures the cross-country relationship, indicating that structurally richer economies are, on average, less prone to systemic crises, consistent with stronger institutions and deeper financial systems providing a buffer against instability. By contrast, current account balances are not significant predictors, while the net financial account is negative and significant in the nonlinear models, consistent with the idea that sudden stops in capital flows raise systemic vul-

[7]The current account balance and the net financial account are linked by balance-of-payments accounting, which can generate collinearity when both are included. However, they are not exact mirror images in the data, since capital account items, statistical discrepancies, and different measurement practices introduce variation. Results are robust to retaining only one of the two measures; for completeness I report specifications with both.As an additional robustness check, I replace the net financial account in levels with its percentage change relative to GDP, capturing shifts in cross-border capital flows rather than stock positions. The results remain qualitatively and quantitatively similar, further reinforcing that the effect of stock market concentration on systemic crisis risk is not sensitive to alternative measures of external imbalances.

nerability. Exchange rate depreciation is positively associated with crisis incidence, while inflation remains insignificant in most cases.

[INSERT Table 6]

Table 7 presents the corresponding average marginal effects (AMEs), which translate the coefficients into probabilities. This effect is economically large and highlights the importance of equity market concentration in the context of systemic financial vulnerability. Domestic credit also exerts a positive marginal effect, though smaller in magnitude, while GDP per capita increases crisis likelihood at the margin. Exchange rate depreciation remains significant, reinforcing its role as an early-warning indicator.

Overall, the baseline results establish three key findings: (i) stock market concentration is a statistically and economically significant predictor of systemic crises; (ii) this association is robust across alternative functional forms; and (iii) macro-financial fundamentals such as domestic credit and exchange rate volatility continue to matter, but their inclusion does not materially alter the relationship between concentration and crisis risk.

The residual term $\hat{v}_{it}$ enters negatively and is statistically significant at the 1% level across all baseline specifications, indicating that stock market concentration is not strictly exogenous in the crisis equation. After accounting for this through the control-function correction, the coefficient on SMC remains positive, economically large, and highly significant, and its magnitude remains stable across specifications. This pattern suggests that the relationship between concentration and systemic crises is not solely driven by reverse causality or omitted variables, and is consistent with stock market concentration remaining robustly associated with systemic vulnerability.

The negative sign on the residual also provides insight into the nature of the endogeneity. It suggests that the unanticipated component of stock market concentration, capturing short-run

deviations from its predicted, structural level, is associated with a lower probability of systemic crisis. One interpretation is that unexpected increases in concentration reflect positive shocks to dominant firms, such as improvements in profitability or favourable global demand conditions, which strengthen aggregate market performance rather than amplify fragility. By contrast, the persistent, structurally determined component of concentration, captured by the main SMC coefficient, remains positively associated with crisis incidence, suggesting that the systemic implications of concentration depend on whether it arises from deep structural features of the market or from transitory shocks.

From an estimation perspective, this negative correlation implies that a naïve specification ignoring endogeneity would underestimate the true positive association between structural concentration and systemic crises, and that the control-function correction reinforces the baseline result.

[INSERT Table 7]

4.2 Baseline Results with Financial Market Controls

Table 8 reports the baseline regression results when stock market depth and liquidity are explicitly controlled for, while Table 9 presents the corresponding average marginal effects. Across all four specifications,probit, logit, cloglog, and LPM,the coefficient on stock market concentration remains positive and statistically significant at the 1% level. This reinforces the central finding that higher concentration in equity markets is robustly associated with a higher probability of systemic crisis.

[INSERT Table 8]

The average marginal effects in Table 9 indicate that a one–standard deviation increase in concentration raises the probability of a systemic crisis by roughly 2–2.6 percentage points across

nonlinear specifications. Given the sample mean crisis incidence of about 10%, this represents a sizeable increase in systemic vulnerability, highlighting the importance of equity market distribution. These estimates align with the argument in Patro et al. (2013) that concentration amplifies systemic risk by increasing correlations and reducing diversification, and with Hou and Robinson (2006), who show that industry concentration shifts return dynamics in ways that magnify firm-specific shocks.

By contrast, the financial depth and liquidity proxies do not exhibit systematic effects on systemic crises. Stock market capitalization is uniformly insignificant across models, while turnover is weakly significant only in the cloglog and LPM specifications, with small economic magnitudes. This suggests that the size and activity of equity markets, while important for intermediation, do not mitigate the risks generated by highly concentrated market structures. This is consistent with Lehkonen (2015), who documents that global market co-movement rose sharply during 2008 regardless of capitalization levels, underscoring that diversification benefits vanish when dominance is concentrated among a limited number of firms.

Control variables behave broadly as expected. Domestic credit enters positively and significantly, consistent with the view that rapid credit expansion raises financial fragility (Laeven and Valencia, 2012; Schularick and Taylor, 2012). Net financial account changes show a negative and significant effect in the cloglog specification, consistent with the idea that sudden reversals in capital inflows are destabilizing. GDP per capita growth enters positively in this specification. This result should not be interpreted as higher growth causing crises; rather, it reflects the inclusion of average GDP levels as a control. Once income levels are conditioned on, the growth coefficient captures short-run accelerations relative to long-run development trends, which in this sample are often associated with credit booms and heightened vulnerability. Inflation remains insignificant,

while external shocks,proxied by exchange rate changes, enter positively and significantly in most models, in line with evidence that sharp depreciations often trigger banking stress (Reinhart and Rogoff, 2009). The residual $\hat{v}_{it}$ remains negative and statistically significant across probit, logit, and cloglog specifications. This pattern is consistent across alternative model specifications that control for stock market capitalization and turnover. These results indicate that the relationship between stock market concentration and systemic crises remains stable after accounting for differences in market size and liquidity conditions.

In aggregate, these results strengthen the paper's main conclusion: stock market concentration is consistently associated with higher systemic crisis risk, even when controlling for financial market depth and liquidity. This finding suggests that concentration in equity markets represents a distinct dimension of systemic vulnerability, rather than merely reflecting broader features of market size or trading activity.

[INSERT Table 9]

4.3 Channel Analysis

Tables 10 and 11 extend the baseline results by incorporating interaction terms between stock market concentration (SMC) and three key dimensions of equity market structure: liquidity (turnover), credit depth, and capitalisation. This allows us to evaluate whether concentration amplifies crisis risk through theoretically grounded channels such as fire-sale contagion, bank fragility, and the stabilising role of market depth. Across all seven specifications, the coefficient on SMC remains positive and statistically significant, reinforcing the robust association between stock market concentration and systemic crises. The average marginal effects in Table 11 indicate that a one–

standard deviation increase in concentration raises the probability of a systemic crisis by roughly 1.7–2.5 percentage points, an effect that is both statistically and economically meaningful given the low unconditional crisis frequency in the sample.

[INSERT Table 10]

[INSERT Table 11]

Liquidity channel. The interaction between SMC and stock market turnover enters positively and is statistically significant across all specifications in which it appears (columns a, d, f, g), with average marginal effects ranging from 2.25×10^{-5} to 2.84×10^{-5}. At the baseline evaluation point of zero turnover, this suggests that concentration amplifies crisis risk through liquidity-driven transmission mechanisms. However, a turnover ratio of zero is not an economically meaningful benchmark in this sample. Evaluating the marginal effect of SMC at the 25th, 50th, and 75th percentiles of the turnover distribution — 16.6, 40.9, and 78.8 respectively — yields estimates of 0.007, 0.007, and 0.005, none of which are statistically distinguishable from zero ($p > 0.34$ throughout). This suggests that while the interaction term captures a statistically detectable pattern, the liquidity channel does not generate robust amplification effects at empirically representative values of market activity. The role of liquidity in transmitting concentration risk therefore remains inconclusive in this cross-country setting, and caution is warranted in drawing strong inferences about fire-sale mechanisms (Coval et al., 2007; Acharya et al., 2011) from the aggregate interaction coefficient alone.

Credit channel. The interaction between SMC and domestic credit is negative but statistically insignificant across all specifications (columns b, d, e, g). At the baseline evaluation point of zero credit, the interaction coefficient offers little interpretable content. Evaluating the marginal effect of SMC at the 25th, 50th, and 75th percentiles of the domestic credit distribution — 40.7,

83.6, and 122.4 percent of GDP respectively — yields estimates of 0.017, 0.026, and 0.031, all of which are highly significant ($p < 0.001$). This pattern indicates that the effect of SMC on crisis probability is economically meaningful and statistically robust across the full range of credit conditions observed in the sample, but does not intensify systematically with credit depth. Equity market concentration thus raises systemic risk independently of the credit environment, rather than compounding the vulnerabilities associated with domestic credit booms. The strong and consistently positive main effect of domestic credit confirms the established link between credit expansions and financial fragility (Schularick and Taylor, 2012; Laeven and Valencia, 2020), but this operates as a parallel rather than an interacting channel. This is broadly consistent with Beck et al. (2006) and Bretschger et al. (2012), who document that bank fragility and equity market structure represent distinct sources of systemic vulnerability.

Market depth channel. The interaction between SMC and stock market capitalization is negative and significant in columns (c), (e), and (f), with average marginal effects ranging from -3.8×10^{-5} to -5.6×10^{-5}. As with the other channels, however, evaluating this interaction at a capitalization ratio of zero provides limited economic insight. Computing the marginal effect of SMC at the 25th, 50th, and 75th percentiles of the capitalization distribution — 31.3, 55.2, and 102.4 percent of GDP respectively — yields estimates of 0.010, 0.019, and 0.034, all statistically significant ($p \leq 0.028$). The rising profile of these estimates indicates that the destabilizing effect of concentration strengthens as equity markets deepen. In larger markets, the systemic footprint of dominant firms expands: their capacity to move prices, absorb liquidity, and transmit shocks to counterparties grows with the overall scale of the market. This finding qualifies the standard view that market depth unconditionally stabilizes the financial system (Hou and Robinson, 2006; Lehkonen, 2015). Rather, depth appears to amplify the systemic relevance of concentrated owner-

ship, suggesting that the protective role attributed to large equity markets is conditional on how capitalization is distributed across firms.

Macroeconomic controls. Control variables behave consistently with theoretical expectations across all specifications. Domestic credit remains a strong and positive predictor of crisis risk, while GDP enters positively and significantly in several columns. Inflation carries a negative coefficient in most specifications, suggesting that deflationary pressures, rather than moderate inflation, may exacerbate systemic fragility. Changes in the exchange rate are strongly positive and significant, consistent with the twin-crisis literature emphasising currency mismatches as crisis triggers (Reinhart and Rogoff, 2009). Current and financial accounts remain weak predictors once concentration and other controls are included, consistent with their complex role in cross-border adjustment dynamics.

In all channel specifications, the residual term $\hat{v}_{it}$ continues to be negative and highly significant, reaffirming the presence of endogeneity. The persistence of this result across specifications that include interactions with liquidity, capitalisation, and credit suggests that these amplification mechanisms operate alongside, rather than substituting for, the baseline relationship between concentration and systemic crises.

In sum, the channel analysis reveals that the systemic implications of stock market concentration are not uniform but vary with the surrounding financial architecture. Evaluated at economically representative values of the moderating variables, the evidence points to a robust and significant amplification effect through market capitalization, an independent and consistently significant direct effect of concentration that does not vary with credit conditions, and an inconclusive liquidity channel whose interaction coefficient does not translate into significant marginal effects at empirically meaningful levels of turnover. These findings underscore that systemic risks in equity

markets cannot be understood by concentration alone, but by its interaction with the scale and structure of the markets in which concentrated firms operate.

5 Robustness

5.1 Alternative Crisis Definition

Systemic crisis episodes are identified using two widely used datasets: Reinhart and Rogoff (2009) and Laeven and Valencia (2020). Because the two sources employ different classification criteria and occasionally identify different crisis episodes, I construct a combined crisis indicator that records a systemic crisis whenever it is identified in either dataset.

Laeven and Valencia (2020) define a systemic banking crisis as an episode in which two conditions are jointly satisfied: (i) the banking system experiences significant distress, such as bank runs, large losses, or widespread bank failures; and (ii) the authorities implement major policy interventions, including bank recapitalizations, nationalizations, blanket guarantees, or large-scale liquidity support (for example exceeding 5% of deposits or liabilities) or fiscal restructuring costs above 3% of GDP. The onset of the crisis is defined as the first year in which both conditions occur.

Formally, the crisis variable equals one if a systemic crisis is reported by Reinhart and Rogoff (2009) or by Laeven and Valencia (2020), and zero otherwise. This approach increases the coverage of crisis episodes and reduces the risk that the results are driven by the specific classification methodology of a single dataset.

Tables 12 and 13 report the results of probit and logit specifications under the Laeven–Valencia

coding. The coefficient on stock market concentration (SMC) remains positive and highly significant across all specifications. In the main specification(Table 12), the SMC coefficient ranges from 0.22 to 0.47 depending on the control set and estimator, with significance at the 1% level in all cases. Average marginal effects (Table 13) show that a one–standard deviation increase in concentration raises systemic crisis probability by 1.6–2.1 percentage points, closely mirroring the magnitudes found under the Reinhart–Rogoff definition.

The robustness specification augments the alternative crisis definition model with all three interaction terms simultaneously, SMC $\times$ Turnover, SMC $\times$ Credit, and SMC $\times$ Capitalisation, to confirm that the main effect of concentration is not absorbed by its interactions with financial structure. SMC remains positive and statistically significant, with a marginal effect ranging from 1.7 to 2.5 percentage points, consistent with the baseline results.

Turning to the interaction terms, the marginal effects evaluated at representative values of each moderating variable reveal a pattern that broadly mirrors the primary channel analysis. The effect of SMC on crisis probability increases with credit depth, rising from 0.005 at P25 to 0.033 at P75, though the estimate at low credit levels is not statistically significant ($p = 0.141$), suggesting that the credit channel activates only at elevated levels of financial intermediation. For market liquidity, the marginal effect increases monotonically from 0.012 to 0.018 across the turnover distribution and remains statistically significant throughout, consistent with fire-sale amplification. For market capitalisation, the effect rises from 0.010 at P25 to 0.034 at P75, all significant, confirming that deeper markets amplify rather than cushion the systemic risk of concentration. These results corroborate the channel findings from the primary analysis and confirm that the conditional relationship between SMC and crisis risk is robust to the use of an alternative crisis definition.

When systemic crises are redefined following Laeven and Valencia (2020), the residual $\hat{v}_{it}$ remains

statistically significant, again indicating that stock market concentration is endogenous in the crisis equation. The robustness of this result across two widely used crisis datasets suggests that the estimated relationship between concentration and systemic crises is not driven by the particular crisis definition employed.[8]

These findings demonstrate that the concentration–crisis link holds consistently across alternative crisis definitions. Instead, it holds under the stricter Laeven–Valencia operationalization, which requires both distress and government intervention. This finding further supports the robustness of the relationship between stock market concentration and systemic crises across alternative datasets.

[INSERT Table 12]

[INSERT Table 13]

5.2 Institutional Variables and Dynamic Lag Specifications

Tables 14 and 15 report robustness checks that examine whether the baseline findings remain stable when additional institutional variables are included and when alternative lag structures for stock market concentration are considered. Specifically, religion and legal origin are incorporated as additional institutional characteristics commonly associated with financial structure in the law-and-finance literature (La Porta et al., 1999; McCaig and Stengos, 2005). In addition, the lag structure of stock market concentration is extended to two and three periods (SMC_{t-2} and SMC_{t-3}) to assess whether the relationship persists when more distant historical values of concentration are considered.

[8] I also estimated the models using the Laeven and Valencia (2020) database, where systemic crises represent about 4% of observations. While the alternative crisis variable combines their definition with Reinhart and Rogoff (2009), I also estimated the models using only the Laeven–Valencia dataset as a robustness check. The results remain qualitatively similar across specifications and are available upon request.

[INSERT Table 14]

Across these specifications, the coefficient on stock market concentration remains positive and statistically significant, with magnitudes broadly comparable to those reported in the baseline results. Average marginal effects in Table 15 indicate that a one-standard-deviation increase in SMC raises crisis probability by roughly 1.7–3.1 percentage points, closely aligned with the baseline estimates. When religion and legal origin are included as additional institutional variables, and when the lag structure of stock market concentration is extended to two and three periods, the residual term $\hat{v}_{it}$ remains negative and statistically significant. This pattern is consistent with the earlier results and indicates that the relationship between stock market concentration and systemic crises remains stable across these alternative specifications.

Taken together, these robustness exercises suggest that the relationship between stock market concentration and systemic crisis risk is not driven by a particular modeling choice. The results remain qualitatively similar when additional institutional variables are included and when deeper lags of concentration are considered. The consistency of the estimates across these alternative specifications strengthens confidence that concentrated equity markets are systematically associated with higher systemic vulnerability.

[INSERT Table 15]

6 Thresholds in Stock Market Concentration and Systemic Stability

The marginal effects plots in Figure 5 provide a more granular view of the mechanisms through which stock market concentration influences systemic crises. Across all specifications, the relationship is non-linear and displays a distinct threshold pattern.

At low levels of concentration (below roughly 25–30%), the estimated marginal effect on crisis probability is close to zero and statistically insignificant. This suggests that moderately concentrated markets do not materially elevate systemic fragility and may even reflect efficient clustering of capital in leading firms.

Beyond this point, however, the crisis risk associated with concentration rises sharply. Between 30% and 60%, the marginal effect increases almost monotonically, indicating that concentration amplifies the transmission of firm-specific shocks and erodes diversification benefits. The effect peaks around 60–70%, where the probability of systemic crisis is estimated to be 1.5–2 percentage points higher relative to markets with moderate concentration. Given that systemic crises are rare but high-impact events, this magnitude is economically significant.

Interestingly, once concentration surpasses roughly 70%, the marginal effect begins to taper off. This pattern reflects the fact that beyond a certain point, additional increments of concentration no longer fundamentally alter the competitive structure of equity markets. When markets have already transitioned to a non-competitive environment dominated by a small set of firms, the systemic risks associated with reduced diversification and distorted capital allocation are already fully at play. In this regime, further concentration adds relatively little to crisis probability compared to the earlier stages where concentration first erodes competition and begins to amplify systemic

fragility.

From a policy perspective, these results highlight the existence of a "danger zone" for stock market concentration. While regulators need not be overly concerned when concentration is below one-third of market capitalization, levels above 50% warrant close monitoring as they consistently signal elevated systemic vulnerability. Incorporating concentration-based thresholds into macroprudential frameworks, for instance, as triggers for heightened disclosure requirements or concentration-adjusted capital surcharges, would allow policymakers to address systemic risks originating outside the traditional banking system. By explicitly recognizing that the destabilizing effects of equity market structure emerge only beyond identifiable thresholds, regulators can better calibrate their interventions to the nonlinear nature of financial fragility.

7 Conclusions

This paper provides new evidence on the systemic consequences of stock market concentration by analyzing a panel of 30 countries from 1998 to 2020. Using multiple nonlinear probability models (probit, logit, cloglog) and complementary linear benchmarks, the results consistently indicate that greater equity market concentration significantly increases the likelihood of systemic financial crises. This effect persists under alternative crisis definitions (Reinhart and Rogoff, 2009; Laeven and Valencia, 2020), across different model specifications (random effects probit, correlated random effects with Mundlak correction, and control-function specifications), and when conditioning on broader measures of financial development and institutional quality.

The empirical analysis highlights how the effect of concentration depends on the surrounding financial structure. First, market depth amplifies rather than mitigates the systemic risk of con-

centration. The marginal effect of concentration rises as market capitalization increases, indicating that the systemic footprint of dominant firms grows with the scale of the market. This qualifies the diversification arguments of Hou and Robinson (2006) and Lehkonen (2015), which suggest that larger markets should dilute the dominance of leading firms. Second, while credit booms independently raise systemic risk (Schularick and Taylor, 2012; Beck et al., 2006), they do not systematically interact with equity concentration, suggesting parallel but distinct channels of fragility. Third, the evidence for a liquidity channel is inconclusive. Although the interaction between concentration and turnover is statistically significant, the implied marginal effects at representative levels of market activity are not, so the results do not provide strong support for fire-sale contagion (Shleifer and Vishny, 2011) as the main mechanism. Overall, these findings underscore that concentration effects are conditional on the surrounding financial architecture. Depth amplifies the risk of concentration, credit booms raise systemic vulnerability through a separate channel, and the role of liquidity remains unresolved.

The econometric robustness of the results further strengthens the argument. Additional specifications incorporating institutional characteristics such as creditor rights (Djankov et al., 2007), legal origin, and religious fractionalization yield results that are consistent with the baseline findings. Similarly, specifications that rely on deeper lags of stock market concentration, following Furceri et al. (2019), Kim and Lin (2011), and McCaig and Stengos (2005), produce qualitatively similar estimates. Across these alternative specifications, the estimated average marginal effect of a one–standard deviation increase in concentration raises systemic crisis probability by roughly 2–3 percentage points, an economically meaningful effect in the context of rare events.

From a policy perspective, the results suggest that the architecture of capital markets is integral to systemic stability. Traditional macroprudential frameworks, which focus primarily on banks and

credit cycles, overlook the vulnerabilities that arise from excessive equity market concentration. The structural nature of stock market concentration is further reinforced by the rise of passive investing. Jiang et al. (2024) show that passive inflows magnify the dominance of large firms through an amplification loop, raising both their valuations and their volatility. This implies that the systemic risk effects documented here are unlikely to dissipate endogenously and call for macroprudential monitoring of equity market structure.

Monitoring concentration as a systemic risk indicator, encouraging broader equity participation, and facilitating access to capital for mid-sized firms are potential policy levers to reduce fragility. Such measures could complement existing banking regulations and provide a more comprehensive approach to financial stability.

Overall, the study contributes to the literature on financial crises by documenting that stock market concentration is not merely a by-product of market evolution, but is closely associated with systemic financial fragility. Its destabilizing impact is amplified in deeper equity markets, conditional on institutional environments, and robust to multiple definitions and econometric approaches. Recognizing equity concentration as a systemic vulnerability is therefore essential for building resilient financial systems in an era of rising dominance by a small number of systemically important firms.

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Data Availability

The data can be provided upon request.

Appendix

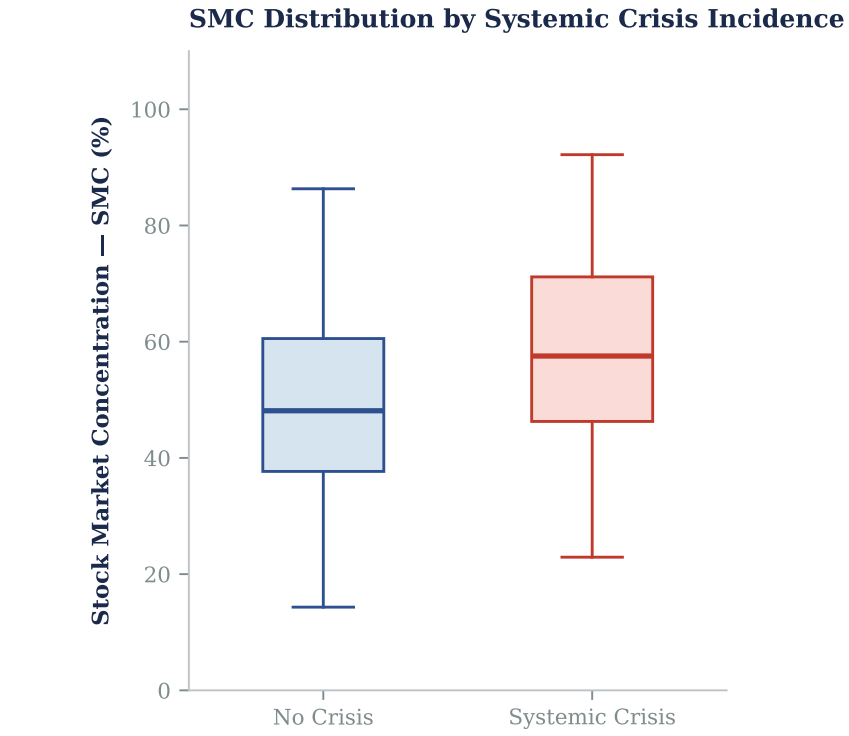


Figure 1: Stock Market Concentration by Crisis Status

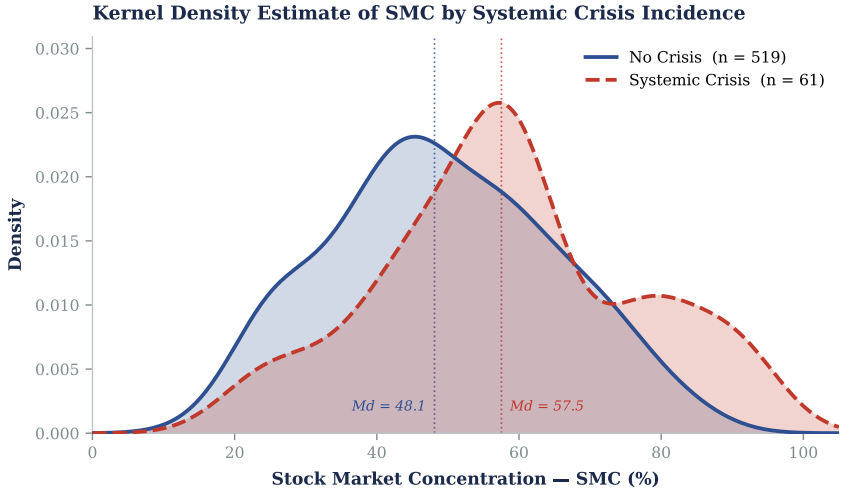


Figure 2: Distribution of Stock Market Concnetration by Crisis

Table 1: Variable Definitions and Data Sources

Variable	**Description**	**Data Source**
Systemic Crisis	Binary indicator equal to 1 if a systemic banking crisis is identified in a given year, 0 otherwise	Harvard BFFS Database
Alternative Indicator	Binary indicator equal to 1 if a systemic banking crisis is identified in a given year, 0 otherwise	Laeven and Valencia (2020)
Stock Market Concentration	Stock Market Concentration, measured as the market capitalization share of the top 10 firms to total domestic stock market capitalization	World Federation of Exchanges
GDP per capita	Annual growth rate of real GDP per capita (%)	WDI, World Bank
Inflation	Inflation rate, measured using the GDP deflator (annual, %)	WDI, World Bank
Δ *Trade*	Change in the trade ratio of international price of exported goods to international price of the imported goods	WDI, World Bank
Δ *Rate*	Change in the real effective exchange rate denominated in USD per local currency	WDI, World Bank
Net Financial Account to GDP	Financial account to nominal GDP ratio	WDI, World Bank
Current Account to GDP	Current account balance as a percentage of GDP	WDI, World Bank
Creditor Rights	Index of creditor protection (0–4), measuring legal rights of lenders against borrowers	Djankov et al. (2007)
Religion Fractionalization	Probability that two randomly selected individuals in a country belong to different religions (ranges 0–1)	Alesina et al. (2003)
Legal Origin	Dummy variables indicating whether a country's legal system is based on English, French, German, or Scandinavian law	La Porta et al. (1998)
Stock Market Capitalization	Stock market capitalization as a percentage of GDP	WDI, World Bank
Stock Market Turnover	Stock market turnover ratio (value traded to market capitalization, %)	WDI, World Bank

Note: This table summarizes all variables used in the empirical analysis, along with their definitions and respective data sources.

Table 2: Aggregate Summary Statistics

Variable	**Mean**	**Std Dev.**	**Min**	**Max**	**Obs**
Systemic Crisis	.1051724	.3070403	0	1	580
Alternative Crisis	.080891	.2728276	0	1	853
Stock Market Concentration	53.11783	18.87451	8.96635	99.09984	853
Domestic Credit	88.01424	50.88892	9.682518	249.2181	660
Current Account	.5298489	6.294516	-16.80081	27.14333	822
Financial Account	.0046614	.0612376	-.1706218	.2644831	819
GDP per capita	23393.52	22084.01	778.5074	112417.9	853
Inflation	4.862268	8.424864	-16.55884	143.6397	853
Δ *Trade*	0.09	2.66	-10.15	17.07	733
Δ *Rate*	.1261263	6.906366	-62.07866	30.0218	607
Stock Market Turnover	257.0376	5723.836	.1469249	165149.1	832
Stock Market Capitalization	93.27032	157.1794	.023168	1777.54	832
Creditor Rights	1.898089	1.055747	0	4	785
Religion Fractionalization	52.00835	37.71545	0.4	99.6	853
Legal Origin	.3622509	.4809328	0	1	853

Note: This table reports summary statistics of the main variables used in the analysis for the 30-country regression sample between 1998 and 2020.

Table 3: Summary Statistics by Country (Part A: A–L)

Country	SysCr	AltCr	SMC	DomCred/GDP	CA/GDP	FA/GDP	GDPpc	Infl	ΔTrade	ΔREER	Turnover	MktCap/GDP	CR	RelFrac	Legor_UK
Australia	0.000	0.000	43.028	111.553	-4.502	-0.045	52017.859	2.664	6.582	1.201	70.298	105.505	3.000	29.800	1.000
Austria	0.000	0.059	62.574	93.344	2.627	0.024	41327.906	1.596	-18.271	-0.074	78.014	25.984	3.000	89.400	0.000
Brazil	0.000	0.000	50.101	44.215	-2.299	-0.021	8048.900	7.420	-30.977	-1.205	59.206	49.191	1.000	87.900	0.000
Canada	0.000	0.000	28.043	109.308	-0.650	-0.006	40635.951	1.946	-96.606	0.524	67.425	121.773	1.000	47.200	1.000
Chile	0.000	0.000	45.159	92.802	-1.077	-0.009	11034.050	4.758	-346.584	-1.291	13.523	98.999	2.000	82.100	0.000
Colombia	0.000	0.000	71.916	33.301	-3.115	-0.030	5481.093	4.579	-20.306	0.245	14.237	45.430	0.000	96.800	0.000
Germany	0.222	0.190	44.752	94.261	4.642	0.038	38419.726	1.243	35.430	-0.629	118.782	47.959	3.000	35.020	0.000
Greece	0.474	0.391	59.255	91.114	-6.318	-0.049	19702.067	1.545	36.632	-0.121	42.627	36.656	1.000	1.900	0.000
Hong Kong		0.000	45.985	169.891	6.750	0.074	35949.451	0.400	6.472	-0.543	52.533	858.970	4.000	8.400	1.000
Hungary		0.063	93.796	46.338	-4.143	-0.031	11244.348	4.917	139.335	1.557	71.995	20.849	1.000	53.900	0.000
Ireland	0.526	0.500	84.320	108.630	-0.519	-0.031	49990.110	2.492	582.687	0.213	19.557	51.798	1.000	95.300	1.000
Iran		0.000	43.879	35.826	5.069	0.043	4607.336	21.936	-80.856	3.212	19.611	54.879	2.000	98.000	0.000
Italy	0.167	0.167	56.759	72.512	-0.638	-0.010	32871.302	2.393	-34.462	0.554	13890.164	40.494	2.000	83.300	0.000
Japan	0.235	0.174	24.817	174.336	2.883	0.030	33344.217	-0.476	209.192	-1.136	103.221	83.389	2.000	0.600	0.000

Note: SysCr = systemic crisis indicator; AltCr = alternative crisis indicator; SMC = top-10 firms' market-cap share; DomCred/GDP = domestic credit to private sector (% GDP); CA/GDP = current account (% GDP); FA/GDP = financial account (% GDP); GDPpc = GDP per capita; Infl = CPI inflation; ΔTrade = change in trade openness; ΔREER = change in real effective exchange rate; Turnover = stock turnover ratio; MktCap/GDP = market capitalization (% GDP); CR = creditor rights; RelFrac = religious fractionalization; Legor_UK = UK legal-origin dummy.

Table 4: Summary Statistics by Country (Part B: M–Z)

Country	SysCr	AltCr	SMC	DomCred/GDP	CA/GDP	FA/GDP	GDPpc	Infl	ΔTrade	ΔREER	Turnover	MktCap/GDP	CR	RelFrac	Legor_UK
Malaysia	0.250	0.190	36.247	116.742	9.085	0.067	8164.218	3.079	-43.209	-0.643	31.928	136.538	3.000	52.200	1.000
Mexico	0.000	0.000	60.604	20.480	-1.517	-0.020	9635.112	6.548	-10.712	-0.752	27.231	29.714	0.000	94.700	0.000
Morocco	0.000	0.000	72.756	83.365	-4.949	-0.046	3047.919	0.677	-21.668	-0.427	6.361	55.372	1.000	99.600	0.000
New Zealand	0.000	0.000	53.998	151.915	-3.715	-0.035	35497.150	2.227	-16.909	0.946	19.597	38.258	4.000	18.700	1.000
Norway	0.000	0.000	69.679	117.390	11.606	0.098	72635.169	3.966	-26.557	-0.301	74.208	53.080	2.000	0.400	0.000
Philippines	0.211	0.190	51.532	33.264	1.599	0.012	2327.716	4.073	2.098	0.911	18.104	56.297	1.000	88.400	0.000
Poland	0.000	0.000	60.141	32.619	-4.069	-0.035	9683.678	3.770	110.795	0.761	38.002	26.530	1.000	81.000	0.000
Russia	0.000	0.091	63.306	47.464	3.904	0.034	9231.964	9.618	14.851	-1.084	40.541	40.464	2.000	12.700	0.000
Singapore	0.000	0.000	40.608	104.657	19.176	0.187	44728.353	0.909	315.913	0.104	50.487	203.915	3.000	22.100	1.000
South Africa	0.000	0.000	34.404	121.021	-2.569	-0.025	5589.294	6.803	139.694	-0.994	28.747	198.333	3.000	11.700	1.000
South Korea	0.313	0.250	40.831	114.821	3.252	0.033	23196.787	2.054	-12.091	2.916	171.649	69.871	3.000	3.900	0.000
Spain	0.000	0.050	49.208	142.273	-3.118	-0.024	25548.897	2.164	68.673	0.323	128.048	76.935	2.000	96.900	0.000
Switzerland	0.000	0.063	68.986	147.933	8.601	0.113	77640.475	0.744	5.055	0.513	69.468	209.471	1.000	53.100	0.000
United Kingdom	.067	0.063	38.318	146.737	-2.459	-0.029	41745.321	2.059	0.091	-1.652	79.701	122.589	4.000	14.500	1.000
United States	.053	0.053	26.844	180.217	-3.570	-0.034	52154.493	1.853	-0.380	-0.006	176.869	128.095	1.000	30.800	1.000

Note: Variable abbreviations as in Table 3.

Table 5: Pearson Correlations among Main Variables

	SysCrisis	AltCrisis	SMC	DomCred/GDP	CurrAcc/GDP	FA/GDP	GDPpc	Inflation	ΔTrade	ΔREER	Turnover	MktCap/GDP	CR	RelFrac	Legor_UK
SysCrisis	1														
AltCrisis	0.9376*	1													
SMC	0.1748*	0.1092*	1												
DomCred/GDP	0.1159*	0.1123*	-0.2887*	1											
CurrAcc/GDP	0.0439	0.0272	-0.0835*	0.1287*	1										
FA/GDP	0.0255	0.0096	-0.0724*	0.3477*	0.9542*	1									
GDPpc	-0.0149	0.0096	0.1438*	-0.0439	0.3477*	0.3452*	1								
Inflation	0.0564	0.0517	0.0287	-0.4591*	-0.0439	-0.0438	-0.2568*	1							
ΔTrade	-0.0153	-0.0097	-0.0229	0.0537	0.0197	0.0215	-0.0314	-0.0105	1						
ΔREER	-0.0517	-0.0119	-0.0062	-0.0416	0.0229	0.0219	0.0026	0.0808*	-0.0269	1					
Turnover	-0.0137	-0.0092	0.0010	0.3662*	0.0063	-0.0039	0.0125	-0.0101	0.0247	0.0359	1				
MktCap/GDP	-0.1878*	-0.0902*	-0.1862*	0.4299*	0.2468*	0.2807*	0.2304*	-0.1132*	-0.0033	0.0393	-0.0207	1			
CR	-0.0327	-0.0396	-0.3943*	0.4790*	0.3125*	0.2965*	0.2767*	-0.0918*	0.0034	-0.0064	0.0049	0.3730*	1		
RelFrac	-0.0282	-0.0308	0.4652*	-0.5020*	-0.2039*	-0.2238*	-0.1667*	0.2194*	-0.0474	0.0190	0.0262	-0.2345*	-0.5529*	1	
Legor_UK	-0.0151	-0.0268	-0.3289*	0.3562*	0.1287*	0.0989*	0.1642*	-0.1651*	0.0515	-0.0190	-0.0268	0.3121*	0.4847*	-0.4612*	1

Note: Pairwise Pearson correlations for all country-year observations. Variables: systemic crisis (SysCrisis), alternative crisis (AltCrisis), stock market concentration (SMC), domestic credit to GDP (DomCred/GDP), current account to GDP (CurrAcc/GDP), financial account to GDP (FA/GDP), GDP per capita (GDPpc), inflation, change in trade openness (ΔTrade), change in real effective exchange rate (ΔREER), stock turnover (Turnover), market capitalization to GDP (MktCap/GDP), creditor rights (CR), religious fractionalization (RelFrac), and UK legal origin dummy (Legor_UK).

Table 6: Regressions for the Effect of Stock Market Concentration on Systemic Risk

	(a) Probit	(b) Logit	(c) Cloglog	(d) LPM
StockMarketConc	0.402***	0.736***	0.456***	0.021***
	(0.083)	(0.165)	(0.076)	(0.007)
Domestic Credit	0.061**	0.111**	0.074***	0.005**
	(0.025)	(0.048)	(0.023)	(0.002)
Current Account	0.044	0.092	-0.019	0.005
	(0.155)	(0.285)	(0.156)	(0.010)
Net Financial Account	-0.199	-0.377	-0.273**	-0.008
	(0.144)	(0.264)	(0.124)	(0.008)
GDP	2.666e-04**	4.994e-04*	2.986e-04*	8.840e-06
	(1.398e-04)	(2.707e-04)	(1.788e-04)	(9.050e-06)
ΔTrade	1.157e-04	2.149e-04	3.050e-05	8.000e-06
	(1.381e-04)	(2.644e-04)	(2.716e-04)	(6.150e-06)
Inflation	-0.028	-0.050	0.028	0.003
	(0.059)	(0.113)	(0.141)	(0.004)
ΔRate	0.080**	0.152**	0.077*	0.002
	(0.039)	(0.077)	(0.039)	(0.002)
Constant	0.882	1.510	0.088	-0.081
	(2.905)	(5.375)	(2.869)	(0.172)
$\hat{v}_{it}$	-0.306***	-0.555***	-0.353***	-0.017***
	(0.059)	(0.116)	(0.074)	(0.005)
σ_u	0.994	1.828		0.140
ρ	0.497	0.504		0.269
Countries	26	26	26	26
Observations	321	321	321	321
(pseudo)R-squared	0.49	0.49	0.57	0.35

Note: This table presents the results of probit, logit, cloglog, and linear probability panel regressions evaluating the effect of stock market concentration on the likelihood of systemic risk. The sample includes 26 countries and 321 observations. All variables are defined in Table 1. Robust standard errors clustered at the country level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 7: Average Marginal Effects of Stock Market Concentration on Systemic Risk

	(a) Probit	(b) Logit	(c) Cloglog	(d) LPM
StockMarketConc	0.024***	0.024***	0.026***	0.021***
	(0.005)	(0.005)	(0.004)	(0.007)
Domestic Credit	0.004***	0.004***	0.004***	0.005**
	(0.001)	(0.001)	(0.001)	(0.002)
Current Account	0.003	0.003	-0.001	0.005
	(0.009)	(0.009)	(0.009)	(0.010)
Net Financial Account	-0.012	-0.012	-0.016**	-0.008
	(0.008)	(0.008)	(0.007)	(0.008)
GDP	1.560e-05***	1.620e-05**	1.710e-05*	8.840e-06
	(6.420e-06)	(7.290e-06)	(9.890e-06)	(9.050e-06)
ΔTrade	6.790e-06	6.970e-06	1.750e-06	8.000e-06
	(7.770e-06)	(8.450e-06)	(1.550e-05)	(6.150e-06)
Inflation	-0.002	-0.002	0.002	0.003
	(0.003)	(0.004)	(0.008)	(0.004)
ΔRate	0.005**	0.005**	0.004**	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Countries	26	26	26	26
Observations	321	321	321	321

Note: This table reports average marginal effects from probit, logit, cloglog, and linear probability models of systemic risk on stock market concentration and control variables. Standard errors clustered at the country level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 8: Regressions for the Effect of Stock Market Concentration on Systemic Risk- Finance Controls

	(a) Probit	(b) Logit	(c) Cloglog	(d) LPM
StockMarketConc	0.395***	0.732***	0.518***	0.020***
	(0.093)	(0.191)	(0.093)	(0.006)
StockMarketCap	0.001	0.000	-0.002	0.001
	(0.011)	(0.021)	(0.013)	(0.000)
StockMarketTurnover	0.008	0.014	0.009*	0.001*
	(0.005)	(0.009)	(0.005)	(0.001)
Domestic Credit	0.067***	0.123**	0.092***	0.005**
	(0.026)	(0.052)	(0.027)	(0.002)
Current Account	0.061	0.123	0.045	0.006
	(0.136)	(0.243)	(0.162)	(0.010)
Net Financial Account	-0.233	-0.452	-0.353**	-0.007
	(0.154)	(0.293)	(0.142)	(0.009)
GDP	2.968e-04*	5.682e-04*	4.485e-04**	5.160e-06
	(1.583e-04)	(3.121e-04)	(1.864e-04)	(9.210e-06)
ΔTrade	8.680e-05	1.543e-04	-2.180e-05	6.960e-06
	(1.270e-04)	(2.309e-04)	(1.635e-04)	(5.960e-06)
Inflation	-0.042	-0.072	-0.022	0.000
	(0.061)	(0.120)	(0.156)	(0.005)
ΔRate	0.077*	0.148	0.090*	0.002
	(0.046)	(0.094)	(0.049)	(0.002)
Constant	4.480	8.502	5.858	-0.279
	(5.392)	(9.145)	(4.609)	(0.183)
$\hat{v}_{it}$	-0.303***	-0.553***	-0.397***	-0.016***
	(0.067)	(0.132)	(0.082)	(0.005)
σ_u	0.667	1.193		0.164
ρ	0.308	0.302		0.347
Countries	26	26	26	26
Observations	319	319	319	319
(pseudo)R-squared	0.53	0.53	0.63	0.37

Note: This table presents the results of probit, logit, cloglog, and linear probability panel regressions evaluating the effect of stock market concentration on the likelihood of systemic risk. The specifications include financial market controls as well as stock market capitalization and stock market turnover. The sample includes 26 countries and 321 observations. All variables are defined in Table 1. Robust standard errors clustered at the country level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 9: Average Marginal Effects of Stock Market Concentration on Systemic Crisis Table-Finance Controls

	(a) Probit	(b) Logit	(c) Cloglog	(d) LPM
StockMarketConc	0.024***	0.024***	0.026***	0.020***
	(0.005)	(0.005)	(0.004)	(0.006)
StockMarketCap	4.020e-05	6.630e-06	-1.113e-04	5.834e-04
	(6.430e-04)	(6.910e-04)	(6.572e-04)	(4.795e-04)
StockMarketTurnover	4.970e-04	4.828e-04	4.523e-04*	0.001*
	(3.183e-04)	(3.222e-04)	(2.558e-04)	(7.751e-04)
Domestic Credit	0.004***	0.004**	0.005***	0.005**
	(0.001)	(0.002)	(0.001)	(0.002)
Current Account	0.004	0.004	0.002	0.006
	(0.008)	(0.008)	(0.008)	(0.010)
Net Financial Account	-0.014	-0.015*	-0.018***	-0.007
	(0.009)	(0.009)	(0.006)	(0.009)
GDP	1.780e-05**	1.900e-05**	2.260e-05***	5.160e-06
	(8.470e-06)	(9.070e-06)	(8.640e-06)	(9.210e-06)
ΔTrade	5.220e-06	5.160e-06	-1.100e-06	6.960e-06
	(7.710e-06)	(7.890e-06)	(8.220e-06)	(5.960e-06)
Inflation	-0.003	-0.002	-0.001	-3.266e-04
	(0.004)	(0.004)	(0.008)	(0.005)
ΔRate	0.005*	0.005*	0.005**	0.002
	(0.003)	(0.003)	(0.002)	(0.002)
Countries	26	26	26	26
Observations	319	319	319	319

Note: This table reports average marginal effects from probit, logit, cloglog, and linear probability models evaluating the effect of stock market concentration and control variables on the likelihood of systemic risk. Standard errors clustered at the country level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 10: Regressions for the Effect of Stock Market Concentration on Systemic Crisis-Interactions

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Stock Market Conc	0.448***	0.462***	0.374***	0.472***	0.388***	0.419***	0.429***
	(0.139)	(0.099)	(0.104)	(0.123)	(0.105)	(0.139)	(0.117)
Stock Market Cap	-0.034**	0.003	0.006	-0.036**	0.007	-0.023	-0.025
	(0.016)	(0.011)	(0.010)	(0.016)	(0.010)	(0.014)	(0.017)
Stock Market Turnover	-0.001	0.009	0.008	0.002	0.008	0.001	0.003
	(0.007)	(0.006)	(0.005)	(0.009)	(0.006)	(0.006)	(0.008)
Domestic Credit	0.104***	0.086**	0.065***	0.121***	0.069**	0.096***	0.105***
	(0.031)	(0.037)	(0.025)	(0.036)	(0.034)	(0.030)	(0.039)
Current Account	0.177	0.009	0.001	0.150	-0.019	0.135	0.125
	(0.157)	(0.105)	(0.139)	(0.139)	(0.088)	(0.147)	(0.140)
Net Financial Account	-0.370***	-0.198	-0.181	-0.342**	-0.165	-0.314	-0.298*
	(0.194)	(0.154)	(0.177)	(0.167)	(0.154)	(0.206)	(0.182)
GDP	5.603e-04***	2.734e-04*	2.269e-04*	5.063e-04***	2.196e-04	4.512e-04***	4.282e-04**
	(1.834e-04)	(1.605e-04)	(1.415e-04)	(1.862e-04)	(1.424e-04)	(1.597e-04)	(1.749e-04)
ΔTrade	-3.270e-06	1.206e-04	-4.930e-05	2.330e-05	-2.900e-05	-1.510e-04	-1.199e-04
	(1.599e-04)	(1.315e-04)	(1.687e-04)	(1.093e-04)	(1.528e-04)	(2.067e-04)	(1.885e-04)
Inflation	-0.091	-0.056	-0.038	-0.113*	-0.038	-0.087*	-0.095*
	(0.066)	(0.050)	(0.042)	(0.062)	(0.040)	(0.052)	(0.056)
ΔRate	0.183***	0.081	0.057	0.191***	0.056	0.153***	0.156***
	(0.060)	(0.052)	(0.042)	(0.059)	(0.042)	(0.049)	(0.048)
Constant	5.267	1.205	-0.862	1.193	-1.832	2.296	0.230
	(3.789)	(7.279)	(2.770)	(5.260)	(3.559)	(2.978)	(5.024)
*StockMarketConc*Turnover*	6.076e-04***			6.950e-04***		5.085e-04***	5.671e-04**
	(1.371e-04)			(2.172e-04)		(1.378e-04)	(2.791e-04)
*StockmarketConc*Credit*		-3.854e-04		-4.457e-04	-9.700e-05		-2.311e-04
		(3.741e-04)		(3.661e-04)	(3.448e-04)		(5.387e-04)
*StockMarketConc*StockMarketCap*			-0.001**		-0.001**	-0.001*	-0.001
			(4.876e-04)		(4.909e-04)	(4.970e-04)	(5.456e-04)
$\hat{v}_{it}$	-0.384***	-0.308***	-0.237***	-0.361***	-0.238***	-0.317***	-0.310***
	(0.107)	(0.067)	(0.076)	(0.101)	(0.077)	(0.097)	(0.094)
σ_u	0.738	0.998	0.551	1.128	0.634	0.667	.006
ρ	0.353	0.499	0.233	0.560	0.287	0.308	0.453
Countries	26	26	26	26	26	26	26
Observations	319	319	391	391	391	391	391
R-squared	0.63	0.54	0.6	0.63	0.6	0.65	0.66

Note: This table reports the results of probit panel regressions evaluating the effect of stock market concentration on the likelihood of systemic financial crises. Column (a) includes the interaction of stock market concentration with Stock Market Turnover, column (b) with Domestic Credit, and column (c) with Stock Market Capitalization, to assess distinct channels through which concentration may affect systemic risk. Columns (d)–(g) combine these interactions. The inclusion of interaction terms allows the effect of concentration to vary with market liquidity, credit depth, and overall market size, highlighting conditional marginal effects. Robust standard errors clustered at the country level are reported in parentheses. All specifications include a Mundlak correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 11: Average Marginal Effects of Stock Market Concentration on Systemic Crisis-Interactions

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Stock Market Conc	0.021***	0.025***	0.020***	0.019***	0.021***	0.019***	0.017***
	(0.006)	(0.004)	(0.004)	(0.006)	(0.003)	(0.005)	(0.006)
Stock Market Cap	-0.002**	0.000	0.000	-0.001**	0.000	-0.001*	-0.001*
	(7.210e-04)	(5.900e-04)	(5.654e-04)	(6.343e-04)	(5.611e-04)	(6.088e-04)	(6.051e-04)
Stock Market Turnover	-3.370e-05	4.933e-04	4.311e-04	9.360e-05	4.320e-04	5.580e-05	1.096e-04
	(3.471e-04)	(3.184e-04)	(3.391e-04)	(3.772e-04)	(3.401e-04)	(2.556e-04)	(3.167e-04)
Domestic Credit	0.005***	0.005***	0.004***	0.005***	0.004**	0.004***	0.004***
	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)
Current Account	0.008	0.001	0.000	0.006	-0.001	0.006	0.005
	(0.007)	(0.006)	(0.008)	(0.005)	(0.005)	(0.007)	(0.005)
Net Financial Account	-0.017***	-0.011	-0.010	-0.014**	-0.009	-0.014*	-0.012*
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.008)	(0.007)
GDP	2.600e-05***	1.490e-05	1.220e-05	2.070e-05**	1.160e-05	2.000e-05***	1.740e-05*
	(8.570e-06)	(9.390e-06)	(7.730e-06)	(8.750e-06)	(7.790e-06)	(7.230e-06)	(9.250e-06)
ΔTrade	-1.520e-07	6.570e-06	-2.660e-06	9.530e-07	-1.530e-06	-6.680e-06	-4.880e-06
	(7.460e-06)	(7.430e-06)	(8.730e-06)	(4.580e-06)	(7.820e-06)	(8.460e-06)	(7.690e-06)
Inflation	-0.004	-0.003	-0.002	-0.005**	-0.002	-0.004**	-0.004**
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
ΔRate	0.009***	0.004*	0.003	0.008***	0.003	0.007***	0.006***
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
*StockMarketConc*Turnover*	2.820e-05***			2.840e-05***		2.250e-05***	2.310e-05***
	(6.320e-06)			(7.380e-06)		(6.840e-06)	(8.410e-06)
*StockmarketConc*Credit*		-2.100e-05		-1.820e-05	-5.140e-06		-9.420e-06
		(2.080e-05)		(1.350e-05)	(1.780e-05)		(2.030e-05)
*StockMarketConc*StockMarketCap*			-5.620e-05***		-5.440e-05**	-3.830e-05**	-3.280e-05
			(2.030e-05)		(2.140e-05)	(1.960e-05)	(2.540e-05)
Countries	26	26	26	26	26	26	26
Observations	319	319	319	319	319	319	319

Note: This table reports average marginal effects (AMEs) from probit panel regressions evaluating the effect of stock market concentration on the likelihood of systemic financial crises. Column (a) includes the interaction of stock market concentration with Stock Market Turnover, column (b) with Domestic Credit, and column (c) with Stock Market Capitalization, to assess distinct channels through which concentration may affect systemic risk. Columns (d)–(g) combine these interactions. AMEs are reported to capture conditional marginal effects, showing how the impact of concentration varies with market liquidity, credit depth, and overall market size. Robust standard errors clustered at the country level are reported in parentheses. All specifications include a Mundlak correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 12: Regressions for the Effect of Stock Market Concentration on Systemic Crisis-Alternative Crisis

	(a)	(b)	(c)	(d)	(e)	(f)
Stock Market Conc	0.253***	0.472***	0.254***	0.468***	0.224***	0.426***
	(0.060)	(0.110)	(0.067)	(0.126)	(0.073)	(0.144)
Stock Market Cap			0.012**	0.021*	0.007*	0.014
			(0.006)	(0.013)	(0.004)	(0.009)
Stock Market Turnover			0.006*	0.012*	0.003	0.006
			(0.003)	(0.006)	(0.003)	(0.005)
Domestic Credit	0.042***	0.078***	0.045***	0.083***	0.047**	0.091**
	(0.014)	(0.026)	(0.015)	(0.028)	(0.022)	(0.045)
Current Account	0.056	0.100	0.072	0.130	0.058	0.106
	(0.079)	(0.151)	(0.080)	(0.158)	(0.072)	(0.137)
Net Financial Account	-0.198***	-0.369***	-0.206***	-0.373***	-0.173**	-0.314**
	(0.069)	(0.135)	(0.068)	(0.135)	(0.071)	(0.131)
GDP	1.706e-04**	3.279e-04**	1.366e-04	2.572e-04	1.034e-04	1.961e-04
	(6.950e-05)	(1.315e-04)	(8.550e-05)	(1.682e-04)	(7.440e-05)	(1.527e-04)
ΔTrade	4.430e-05	7.950e-05	3.690e-05	5.730e-05	2.960e-05	5.770e-05
	(6.330e-05)	(1.312e-04)	(7.040e-05)	(1.486e-04)	(6.410e-05)	(1.237e-04)
Inflation	0.082	0.139	0.075	0.147	0.084	0.167
	(0.058)	(0.119)	(0.064)	(0.137)	(0.062)	(0.143)
ΔRate	0.045**	0.087**	0.040*	0.070	0.052**	0.093**
	(0.023)	(0.043)	(0.023)	(0.048)	(0.024)	(0.049)
Constant	-2.874*	-4.814	-3.620**	-6.105*	-4.825	-9.247
	(1.650)	(3.043)	(1.820)	(3.740)	(3.742)	(7.702)
*StockMarketConc*Turnover*					1.606e-04**	2.870e-04*
					(8.140e-05)	(1.718e-04)
*StockmarketConc*Credit*					-1.287e-04	-3.114e-04
					(2.936e-04)	(5.763e-04)
*StockMarketConc*StockMarketCap*					-2.830e-04	-5.508e-04
					(2.045e-04)	(3.950e-04)
$\hat{v}_{it}$	-0.207***	-0.388***	-0.205***	-0.379***	-0.162***	-0.304***
	(0.054)	(0.097)	(0.059)	(0.110)	(0.052)	(0.105)
σ_u	0.678	1.208	0.666	1.156	0.898	1.755
ρ	0.315	0.307	0.307	0.289	0.446	0.484
Countries	30	30	30	30	30	30
Observations	402	402	399	399	399	399
R-squared	0.36	0.36	0.41	0.41	0.47	0.47

Note: This table reports the results of panel probit and logit regressions evaluating the effect of stock market concentration on the likelihood of systemic financial crises. Systemic crises are defined using the BFFS dataset and complemented with Laeven and Valencia's systemic crisis episodes. Columns (a), (c), and (e) present probit specifications, while columns (b), (d), and (f) report corresponding logit models. Robust standard errors clustered at the country level are reported in parentheses. All specifications include a Mundlak correction to account for unobserved heterogeneity. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 13: Average Marginal Effects of Stock Market Concentration on Systemic Risk-Alternative Crisis

	(a)	(b)	(c)	(d)	(e)	(f)
Stock Market Conc	0.021***	0.021***	0.020***	0.020***	0.016***	0.016***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Stock Market Cap			9.575e-04*	8.760e-04*	5.425e-04*	5.290e-04
			(4.920e-04)	(5.354e-04)	(3.061e-04)	(3.443e-04)
Stock Market Turnover			4.900e-04	4.835e-04*	1.913e-04	2.224e-04
			(2.726e-04)	(2.699e-04)	(2.162e-04)	(1.970e-04)
Domestic Credit	0.004***	0.004***	0.004***	0.003***	0.003***	0.004**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Current Account	0.005	0.005	0.006	0.005	0.004	0.004
	(0.006)	(0.007)	(0.006)	(0.007)	(0.005)	(0.005)
Net Financial Account	-0.017***	-0.017***	-0.016***	-0.016***	-0.013***	-0.012**
	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)
GDP per capita	1.440e-05***	1.500e-05***	1.060e-05*	1.080e-05*	7.510e-06	7.580e-06
	(5.480e-06)	(5.590e-06)	(6.240e-06)	(6.690e-06)	(5.630e-06)	(6.050e-06)
ΔTrade	3.450e-06	3.290e-06	2.860e-06	2.400e-06	2.150e-06	2.230e-06
	(5.290e-06)	(6.100e-06)	(5.490e-06)	(6.260e-06)	(4.780e-06)	(4.840e-06)
Inflation	0.007	0.006	0.006	0.006	0.006	0.006
	(0.005)	(0.006)	(0.005)	(0.006)	(0.005)	(0.006)
ΔRate	0.004**	0.004**	0.003*	0.003	0.004**	0.004**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
*StockMarketConc*Turnover*					1.170e-05**	1.110e-05*
					(5.510e-06)	(6.030e-06)
*StockMarketConc*Credit*					-9.340e-06	-1.200e-05
					(2.050e-05)	(2.120e-05)
*StockMarketConc*StockMarketCap*					-2.050e-05	-2.130e-05
					(1.380e-05)	(1.430e-05)
Countries	30	30	30	30	30	30
Observations	402	402	399	399	399	399

Note: This table reports average marginal effects (AMEs) corresponding to the panel probit and logit specifications in Table 12. Systemic crises are defined using the BFFS dataset and complemented with Laeven and Valencia's systemic crisis episodes. Robust standard errors clustered at the country level are reported in parentheses. All specifications include a Mundlak correction to account for unobserved heterogeneity. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 14: Regressions for the Effect of Stock Market Concentration on Systemic Crisis-Robustness

	(a)	(b)	(c)	(d)	(e)	(f)
Stock Market Conc	0.395***	0.429***	0.651***	0.755***	0.620***	0.562***
	(0.093)	(0.117)	(0.126)	(0.142)	(0.182)	(0.205)
Stock Market Cap	6.688e-04	-0.025	0.011	-0.034	0.002	-0.040***
	(0.011)	(0.017)	(0.010)	(0.021)	(0.009)	(0.016)
Stock Market Turnover	0.008	0.003	0.010**	0.002	0.008**	0.001
	(0.005)	(0.008)	(0.005)	(0.010)	(0.004)	(0.005)
Domestic Credit	0.067**	0.105***	0.096***	0.136**	0.066**	0.072***
	(0.026)	(0.039)	(0.028)	(0.053)	(0.028)	(0.027)
Current Account	0.061	0.125	0.301*	0.398***	0.047	-0.135
	(0.136)	(0.140)	(0.161)	(0.122)	(0.181)	(0.168)
Net Financial Account	-0.233	-0.298*	-0.443*	-0.583***	-0.257	-0.202
	(0.154)	(0.182)	(0.241)	(0.209)	(0.231)	(0.239)
GDP	2.968e-04*	4.282e-04**	4.134e-04**	6.255e-04***	4.560e-04**	6.801e-04
	(1.583e-04)	(1.749e-04)	(1.702e-04)	(1.645e-04)	(1.866e-04)	(2.348e-04)
ΔTrade	8.680e-05	-1.199e-04	9.840e-06	-1.629e-04	1.434e-04*	6.150e-05
	(1.270e-04)	(1.885e-04)	(1.741e-04)	(3.083e-04)	(7.980e-05)	(1.334e-04)
Inflation	-0.042	-0.095*	-0.066	-0.070	-0.129	-0.587*
	(0.061)	(0.056)	(0.077)	(0.053)	(0.114)	(0.310)
ΔRate	0.077*	0.156***	0.050	0.178***	0.019	0.086***
	(0.046)	(0.048)	(0.044)	(0.074)	(0.041)	(0.034)
Constant	4.480	0.230	5.286	0.515	5.500	3.467
	(5.392)	(5.024)	(6.030)	(4.926)	(4.510)	(4.303)
*StockMarketConc*Turnover*		5.671e-04**		8.221e-04**		6.484e-04***
		(2.791e-04)		(3.653e-04)		(1.341e-04)
*StockmarketConc*Credit*		-2.311e-04		-1.901e-04		-1.324e-04
		(5.387e-04)		(6.241e-04)		(2.412e-04)
*StockMarketConc*StockMarketCap*		-8.051e-04		-7.687e-04		-6.501e-04
		(5.456e-04)		(8.688e-04)		(5.707e-04)
$\hat{v}_{it}$	-0.303***	-0.310***	-0.490***	-0.634***	-0.464***	-0.429
	(0.067)	(0.094)	(0.096)	(0.135)	(0.164)	(0.182)
σ_u	0.667	0.909	1.137	1.377	0.638	5.419e-04
ρ	0.308	0.453	0.564	0.655	0.289	2.940e-07
Countries	26	26	26	26	26	26
Observations	319	319	319	319	303	303
R-squared	0.53	0.66	0.58	0.7	0.58	0.69

Note: This table presents robustness checks from panel probit regressions evaluating the relationship between stock market concentration and systemic financial crises. Columns (a)–(b) augment the baseline specification with additional institutional variables (religious fractionalization and legal origin). Columns (c)–(d) extend the lag structure of the explanatory variables from X_{t-1} to X_{t-2}, while columns (e)–(f) further extend it to X_{t-3} while maintaining the additional institutional variables. Robust standard errors clustered at the country level are reported in parentheses. All models include a Mundlak correction to account for unobserved heterogeneity. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Variable definitions are provided in Table 1.

Table 15: Average Marginal Effects of Stock Market Concentration on Systemic Risk — Robustness

	(a)	(b)	(c)	(d)	(e)	(f)
Stock Market Conc	0.024***	0.017***	0.030***	0.024***	0.031***	0.025***
	(0.005)	(0.006)	(0.005)	(0.004)	(0.006)	(0.004)
Stock Market Cap	4.020e-05	-0.001*	5.205e-04	-0.001*	1.021e-04	-0.002***
	(6.430e-04)	(6.051e-04)	(5.027e-04)	(5.711e-04)	(4.623e-04)	(5.622e-04)
Stock Market Turnover	4.970e-04	1.096e-04	4.796e-04*	6.690e-05	4.190e-04*	3.420e-05
	(3.183e-04)	(3.167e-04)	(2.704e-04)	(2.994e-04)	(2.570e-04)	(2.267e-04)
Domestic Credit	0.004***	0.004***	0.004***	0.004***	0.003**	0.003**
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)
Current Account	0.004	0.005	0.014**	0.013***	0.002	-0.007
	(0.008)	(0.005)	(0.007)	(0.004)	(0.009)	(0.007)
Net Financial Account	-0.014	-0.012*	-0.020**	-0.018**	-0.013	-0.008
	(0.879)	(0.732)	(0.984)	(0.764)	(0.972)	(0.830)
GDP	1.780e-05**	1.740e-05*	1.890e-05***	1.970e-05***	2.260e-05***	2.940e-05***
	(8.470e-06)	(9.250e-06)	(6.890e-06)	(5.950e-06)	(8.040e-06)	(5.800e-06)
ΔTrade	5.220e-06	-4.880e-06	4.500e-07	-5.120e-06	7.120e-06	3.470e-06
	(7.710e-06)	(7.690e-06)	(7.990e-06)	(1.020e-05)	(4.730e-06)	(5.610e-06)
Inflation	-0.003	-0.004**	-0.003	-0.002*	-0.006	-0.025***
	(0.004)	(0.002)	(0.003)	(0.001)	(0.006)	(0.009)
ΔRate	0.005*	0.006***	0.002	0.006***	0.001	0.003**
	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
StockMarketConc × Turnover		2.310e-05***		2.590e-05***		2.860e-05***
		(8.410e-06)		(9.170e-06)		(5.490e-06)
StockMarketConc × Credit		-9.420e-06		-5.980e-06		-5.510e-06
		(2.030e-05)		(1.890e-05)		(1.010e-05)
StockMarketConc × Stock Mkt Cap		-3.280e-05		-2.420e-05		-2.550e-05
		(2.540e-05)		(2.890e-05)		(1.800e-05)
Countries	26	26	26	26	26	26
Observations	399	399	399	399	399	399

Note: Table 15 reports average marginal effects (AMEs) corresponding to the robustness specifications in Table 14. AMEs are computed from panel probit models of systemic financial crises (BFFS, complemented with Laeven–Valencia). Columns (a)–(b) include additional institutional variables (religious fractionalization and legal origin); columns (c)–(d) and (e)–(f) extend the lag structure of the explanatory variables to X_{t-2} and X_{t-3}, respectively. AMEs are interpretable as percentage–point changes in crisis probability. Robust standard errors clustered at the country level are reported in parentheses. All models include a Mundlak correction to account for unobserved heterogeneity. *Significance:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: First-Stage Predetermined Factor Relevance and Diagnostics

	(a)	(b)	(c)	(d)	(e)	(f)
Panel A: First stage (dependent variable: SMC_{it})						
Creditor Rights	3.783***	3.675***	4.194***	4.388***	5.331***	7.019***
	(1.160)	(1.391)	(1.448)	(1.301)	(1.386)	(1.471)
Religion			1.038**	1.210**	1.178**	1.605**
			(0.506)	(0.485)	(0.570)	(0.657)
Legal Origin			-16.920***	-17.456***	-21.715***	-27.435***
			(3.990)	(4.129)	(3.827)	(4.111)
Panel B: First-stage diagnostics (matching each column/spec)						
Joint F-statistic (predetermined factors)	30.830***	17.880***	107.750***	240.520***	134.120***	64.980***
Observations	321	321	321	321	321	321
Country FE / Year FE	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes

Note: Panel A reports coefficients on the predetermined factors from the first-stage regression of stock market concentration (SMC) on its own lag, control variables, and predetermined factors, with cluster-robust standard errors in parentheses. Column (a) is the baseline; (b) adds financial controls; (c)–(d) replicate with the full set of predetermined factors; (e)–(f) shift the autoregressive component to X_{t-2} and X_{t-3}. Panel B reports the joint F-statistic of the predetermined factors, testing their collective relevance in explaining variation in stock market concentration. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 17: Diagnostics: Random Effects vs. Mundlak Specifications

	(1) Baseline	(2) Finance	(3) Interactions	(4) Alt. Crisis	(5) Add. Instr.	(6) Lagged SMC
LM test (H_0: $\rho = 0$)						
χ^2	32.29	30.22	51.95	58.64	30.42	55.02
p-value	0.0001	0.0008	0.0000	0.0000	0.0007	0.0000
Wald test: Mundlak means joint						
χ^2	40.41	35.68	35.79	23.77	23.77	29.43
p-value	0.0000	0.0001	0.0001	0.0082	0.0082	0.0011

Note: Specification diagnostics comparing standard random effects and Mundlak (correlated random effects) panel probit models. The upper panel reports LM tests of $H_0 : \rho = 0$ (no random effects), which are highly significant across columns (1)–(6), indicating non-negligible unobserved heterogeneity. The lower panel reports joint Wald tests of the Mundlak means; significance in all cases supports including Mundlak terms to mitigate correlation between regressors and unit effects. Results are consistent across additional model variants; for brevity, only the main configurations are shown.

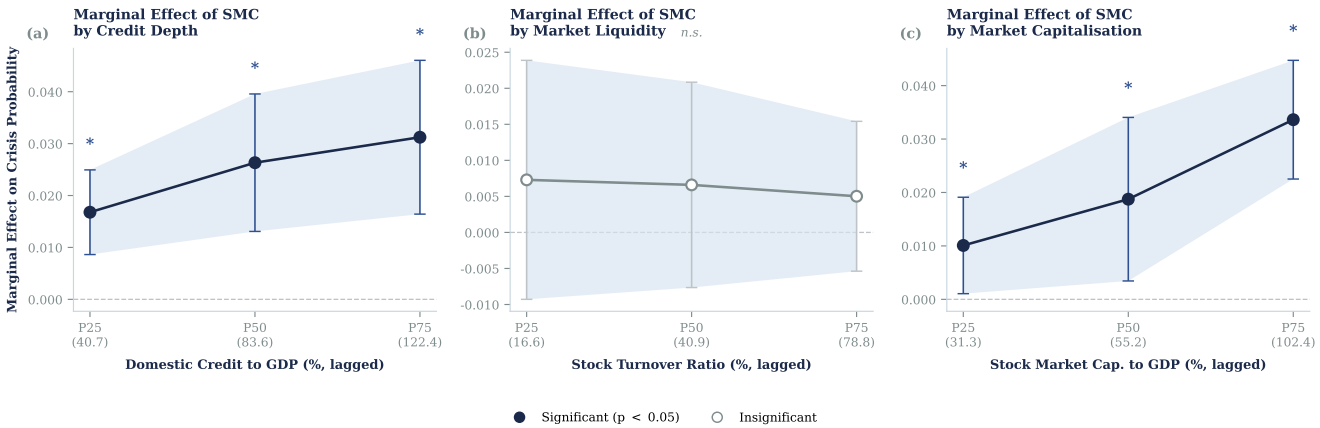


Figure 3: Conditional Marginal Effects of SMC on Systemic Crisis Probability at Representative Values of Moderating Variables (Baseline Definition). Notes: * denotes significance at the 5% level. Marginal effects evaluated at the 25th, 50th, and 75th percentiles of the moderating variable. Crisis years defined following Reinhart & Rogoff (2009), Harvard BFFS dataset. Random-effects probit with Mundlak correction and control-function residual. Robust standard errors (Delta method).

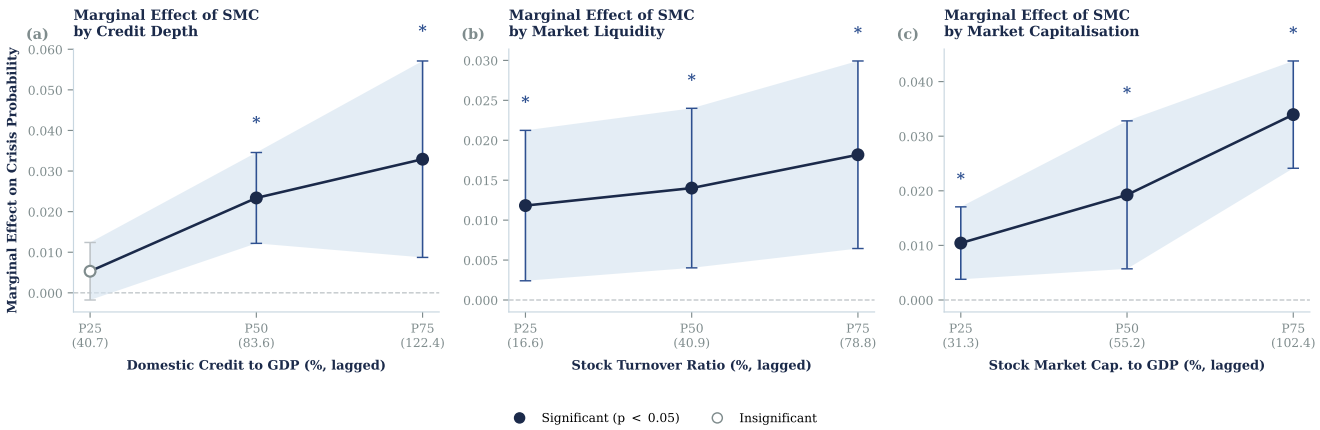
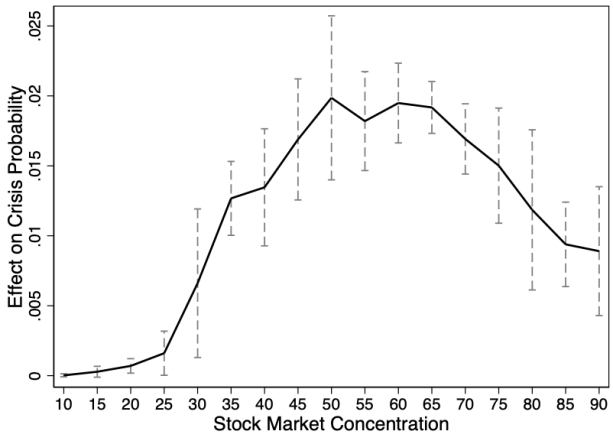
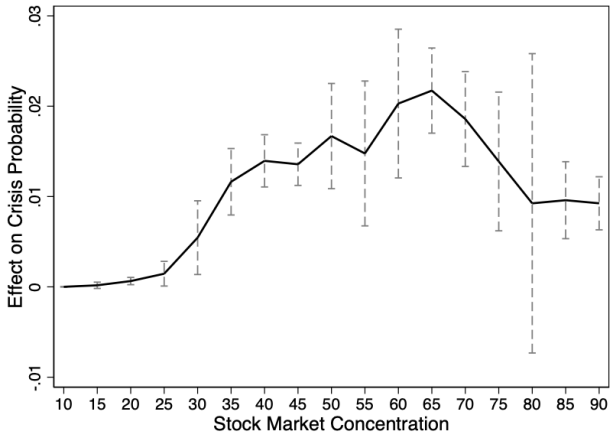


Figure 4: Conditional Marginal Effects of SMC on Crisis Probability at Representative Values of Moderating Variables (Alternative Crisis Definition). Notes: * denotes significance at the 5% level. Marginal effects evaluated at the 25th, 50th, and 75th percentiles of the moderating variable. The alternative crisis definition combines systemic and currency crisis episodes. Random-effects probit with Mundlak correction and control-function residual. Robust standard errors (Delta method).

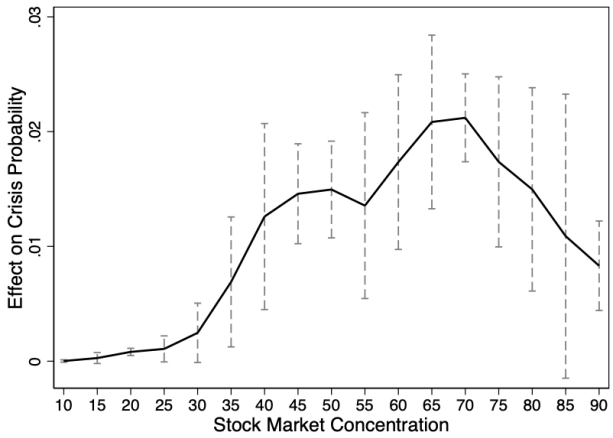
Marginal Effects of Stock Market Concentration on Systemic Crisis



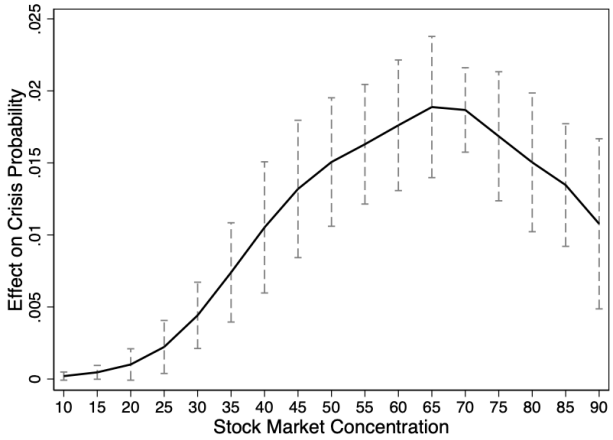
(a) Baseline probit



(b) Baseline with Finance Controls



(c) Interactions Probit



(d) Alternative Crisis

Figure 5: Marginal effects of stock market concentration on the probability of systemic crisis under alternative model specifications. Dashed lines denote 95% confidence intervals.